



Final Study Report

Evaluation of Light Transmission and Polymerization of Varied Resin Composites Through Solventum™ Filtek™ Matrix at Variable Thicknesses

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List of figures: Figure 1 (common-scale beam panel); Figure 2 (material-minus-air difference maps); Figure 3 (emission spectra of the four LPUs, normalized and absolute); Figure 4 (wavelength-resolved transmission through each material); Figure S1 (BeamGage screenshots of the same conditions as Figure 1); Figure S2 (real-time conversion traces with light status).

Abbreviations and terminology. B/T, bottom-to-top hardness ratio. CV, coefficient of variation. $D4\sigma$, second-moment beam diameter (ISO 11145 and 11146-1): four times the standard deviation of the irradiance distribution about the beam centroid, evaluated separately in X and Y, so the beam width reflects how all of the light is spread rather than where a visible edge falls. DC, degree of conversion. EXA, EXACLEAR. FM, Filtek Matrix. FTIR-ATR, Fourier transform infrared spectroscopy with attenuated total reflectance. KHN, Knoop hardness number. LPU, light-polymerization unit. SD, standard deviation. Terminology follows The Journal of Prosthetic Dentistry and the Journal of Dentistry: "polymerize" rather than "cure," "specimen" rather than "sample," "light-polymerization unit" rather than "curing light." Filtek Matrix is a custom 3D-printed clear polymer matrix; EXACLEAR is a clear vinyl polysiloxane index; together they are referred to as light-transmitting matrix materials.

1. Executive summary

Four light-polymerization units (LPUs) were characterized through a custom 3D-printed clear polymer matrix (Filtek Matrix, 1 and 2 mm) and a clear vinyl polysiloxane index (EXACLEAR, 5 and 10 mm) by spectroradiometry, laser beam profiling, FTIR degree of conversion, and Knoop hardness. Three findings carry the report.

First, Filtek Matrix transmits blue light almost completely, 96 to 104% of the matched air value for all four LPUs, and attenuates violet light substantially for multi-peak units: VALO X to 31 to 56% and Bluephase PowerCure to 58 to 77% at 1 to 2 mm. Single-peak units are unaffected.

Second, neither matrix material changes where light is delivered. Beam diameter, shape, uniformity, and hot-spot positions are preserved through both materials. The matrix behaves as a weak diffuser that softens edges and redistributes a small fraction of light to the beam margin.

Third, neither matrix material changes the outcome in the resin composite. Degree of conversion and top and bottom Knoop hardness through the matrix were statistically indistinguishable from direct-contact polymerization for both resin composites and both LPUs tested, including the multi-peak unit with the resin composite most responsive to violet. The study could detect a loss of 1 to 2 conversion points, so the conversion result is a firm negative; hardness was slightly less sensitive

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(Section 11). **Conversion and hardness were governed by the resin composite and its match to the emission spectrum of the LPU, not by the interposed material.** Polymerizing through Filtek Matrix at clinically relevant thickness requires no exposure compensation with either LPU tested in Part 2 (Elipar DeepCure-S and VALO X), and the blue transmission data indicate the same for SmartLite Pro and Bluephase PowerCure, provided the tip is placed on the matrix; air distance costs more irradiance than the material does.

Key results

Result	Value	Source
Blue light (425 to 550 nm) transmitted through Filtek Matrix at 1 to 2 mm, all four LPUs	96 to 104% of matched air	Table 2
Violet light (350 to 425 nm) transmitted through Filtek Matrix, multi-peak LPUs	31 to 77% of matched air	Table 2
Change in beam diameter ($D4\sigma$) through either material	2% larger to 4% smaller than air	Table 8
Degree of conversion and Knoop hardness through either material compared with direct contact	No difference, all Dunnett $P > .05$	Tables 12 to 15
Irradiance lost to 10 mm of air compared with irradiance lost to the materials	54 to 70% (air) compared with 0 to 20% (materials)	Tables 1, 2, 3, 10

2. Objectives and status

Objective (as proposed)	Status	Where reported
1. Quantify wavelength-specific and total irradiance transmission through Filtek Matrix (1.0 and 2.0 mm) and EXACLEAR (5.0 and 10.0 mm)	Met, with change of instrument	Section 1A, Tables 1 to 4, Figure 4
2. Characterize beam profile and spatial energy distribution after transmission	Met, one frame per condition	Section 1B, Tables 5 to 10, Figures 1 and 2
3. Evaluate degree of conversion at the bottom surface of resin composite beneath the matrix materials	Partly met: thin-film DC rather than bottom-surface DC	Section 2A, Table 12
4. Assess bottom-surface microhardness as a proxy for polymerization quality	Met and extended: Knoop on top and bottom, with B/T ratio	Section 2B, Tables 13 to 16
5. Compare single-peak and multi-peak units	Met	Sections 1A, 1C, 6; Figure 3

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6. Determine polymerization recommendations for varied matrix thicknesses	Met	Section 8
7. Compare resin composites with camphorquinone and with alternative photoinitiator systems	Met	Sections 2A to 2C, 6

3. Materials, instruments, and software

Table M1. Light-polymerization units

LPU	Manufacturer	Emission	Serial	Radiant power at contact (mW)	Radiant exitance at tip (mW/cm ²)	Emission peak (nm)	Violet fraction of output (385 to 425 nm)
Elipar DeepCure-S	Solventum	Single-peak	Serial 933112000964	715	911	449	1.2%
SmartLite Pro	Dentsply Sirona	Single-peak, standard tip	Ref K3344400, serial H00050	983	1252	464	0.2%
VALO X	Ultradent	Multi-peak	Ref 5973, serial 3F001066	1313	1070	447	25.2%
Bluephase PowerCure	Ivoclar	Multi-peak	Ref 667092BU, serial 1428009282	569	895	446	17.1%

Radiant power, radiant exitance (radiant power divided by tip emitting area), peak, and violet fraction from integrating sphere spectroradiometry (Table M3), 10 mm aperture, May 20, 2026, mean of five readings. A second session on March 25, 2026 with a 7 mm aperture gave 729, 999, 1362, and 582 mW, within 2% of the May values. Emission spectra recorded with the cosine-corrector spectroradiometer are shown in Figure 3.

Table M2. Matrix materials and resin composites

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Material	Manufacturer	Lot	Form used
Filtek Matrix	Solventum	Provided by Solventum	3D-printed flat plates, 20 × 20 mm, 1.0 and 2.0 mm
EXACLEAR	GC	Ref 008482, lot 2506021	20 × 20 mm, 5.0 and 10.0 mm
Filtek Supreme Ultra, A2 Body	Solventum	Ref 6029A2B, lot 13483783	Thin films (DC); 5 × 2 mm disks (hardness)
Vit-l-escence, Universal A2	Ultradent	Ref UP-1202, lot C7WH4	Thin films (DC); 5 × 2 mm disks (hardness)

Table M3. Instruments and software

Item	Model	Manufacturer	Serial	Additional information
Spectroradiometer with cosine corrector	Ocean Optics STS, config STS-VIS-L-100-400-SMA with CC-3-DA	Ocean Optics	S07815	Part 030-00000-209; transmission through the materials (Section 1A)
Integrating sphere spectroradiometer	Ultradent Minigig	Gigahertz-Optik	45047	Software 2019.1.0, firmware 1.64; radiant power and spectra of each LPU at contact; reference for profiler power calibration
Laser beam profiler	BGP-USB3-SP932U with BeamGage Professional 6.25	Ophir-Spiricon	23110249	Power calibrated against the integrating sphere radiant power, May 20, 2026
FTIR spectrometer with ATR	Invenio R with OPUS 8.5 SP1 (build 8.7.10)	Bruker	391	Kinetic acquisition (CHROM), external photodetector channel for light activation
Microhardness tester	HM-220, Knoop indenter	Mitutoyo	000142212	Code 810-409-11; built 2022
Statistical software	Prism 11.0.0	GraphPad		Analyses independently reproduced in Python 3

				(statsmodels, SciPy); results agree
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4. Part 1. Light transmission characterization

4.1 Section 1A. Spectral irradiance transmission

Method. Irradiance was recorded with the cosine-corrector spectroradiometer and integrated over 350 to 550 nm (total), 425 to 550 nm (blue), and 350 to 425 nm (violet). The tip was placed directly on the cosine corrector at 0 mm; air gaps of 1, 2, 5, and 10 mm were set with spacers on an optical breadboard. Material specimens were held in direct contact with the cosine corrector, so the exit face of each material was index coupled to the sensor. Five readings were made per condition (4 LPUs $\times$ 9 conditions $\times$ 5 = 180 measurements). Two-way ANOVA (LPU $\times$ condition) was performed per band and thickness with the Tukey honestly significant difference test ($\alpha=.05$). Transmission is expressed as the material value divided by the air value at the same tip-to-sensor distance.

Deviation from the proposal. An integrating sphere was proposed for the transmission measurements; a cosine-corrector spectroradiometer was used because the clear films would not rest in direct contact with the sensor, and the integrating sphere was used for contact radiant power and spectra (Table M1). The cosine-corrector sensor is smaller than the beam, so readings at distance reflect the beam overfilling the sensor as well as true divergence; Section 1C addresses the consequence.

Data note. An early data file carried duplicated DeepCure-S entries through EXACLEAR; the values reported here are from the verified final dataset.

Table 1. Irradiance in air at five tip-to-sensor distances (mW/cm^2 , mean $\pm$ SD, $n=5$). Uppercase letters: units within a distance (column). Lowercase letters: distances within a unit (row). Tukey test, $\alpha=.05$.

Total 350 to 550 nm

LPU	0 mm	1 mm	2 mm	5 mm	10 mm
DeepCure-S	1926.7 $\pm$ 13.1 Aa	1817.4 $\pm$ 19.6 Ab	1690.0 $\pm$ 26.9 Ac	1264.4 $\pm$ 19.1 Ad	885.7 $\pm$ 6.7 Ae
SmartLite Pro	1338.6 $\pm$ 39.7 Ca	1174.4 $\pm$ 7.7 Cb	1041.3 $\pm$ 10.0 Dc	830.1 $\pm$ 6.7 Dd	575.7 $\pm$ 11.1 Be
VALO X	1507.2 $\pm$ 7.3 Ba	1310.5 $\pm$ 10.6 Bb	1187.0 $\pm$ 9.5 Cc	871.0 $\pm$ 3.1 Cd	563.6 $\pm$ 3.0 Be

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Bluephase PowerCure	1368.9 ± 0.6 Ca	1309.0 ± 4.9 Bb	1268.3 ± 1.5 Bc	926.3 ± 12.0 Bd	417.0 ± 12.4 Ce
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Blue 425 to 550 nm

LPU	0 mm	1 mm	2 mm	5 mm	10 mm
DeepCure-S	1896.2 ± 12.9 Aa	1788.5 ± 19.3 Ab	1663.1 ± 26.5 Ac	1247.0 ± 18.8 Ad	874.2 ± 6.5 Ae
SmartLite Pro	1331.6 ± 39.4 Ba	1168.3 ± 7.7 Bb	1035.9 ± 9.9 Bc	825.8 ± 6.6 Bd	572.8 ± 11.0 Be
VALO X	1084.0 ± 4.7 Da	963.1 ± 6.3 Db	887.2 ± 6.2 Cc	672.5 ± 2.2 Dd	438.1 ± 2.2 Ce
Bluephase PowerCure	1141.2 ± 0.5 Ca	1093.2 ± 3.8 Cb	1061.4 ± 1.5 Bc	781.6 ± 9.2 Cd	354.3 ± 11.6 De

Violet 350 to 425 nm

LPU	0 mm	1 mm	2 mm	5 mm	10 mm
DeepCure-S	30.5 ± 0.3 Ca	28.9 ± 0.4 Cb	26.9 ± 0.5 Cc	17.4 ± 0.3 Cd	11.6 ± 0.1 Ce
SmartLite Pro	7.0 ± 0.3 Da	6.1 ± 0.0 Db	5.4 ± 0.1 Dc	4.3 ± 0.1 Dd	3.0 ± 0.1 De
VALO X	423.2 ± 2.6 Aa	347.4 ± 4.3 Ab	299.8 ± 3.3 Ac	198.6 ± 1.0 Ad	125.4 ± 0.7 Ae
Bluephase PowerCure	227.7 ± 0.2 Ba	215.8 ± 1.2 Bb	206.9 ± 0.7 Bc	144.7 ± 3.0 Bd	62.6 ± 5.2 Be

Table 2. Irradiance through Filtek Matrix and through the matched air gap (mW/cm², mean ± SD, n=5). * differs from air within the same LPU (Tukey test, P<.05). T = transmission relative to air.

Total 350 to 550 nm

LPU	Air 1 mm	FM 1 mm	T (%)	Air 2 mm	FM 2 mm	T (%)
DeepCure-S	1817.4 ± 19.6	1762.3 ± 25.9 *	97.0	1690.0 ± 26.9	1655.3 ± 8.6 *	97.9
SmartLite Pro	1174.4 ± 7.7	1169.2 ± 21.0	99.6	1041.3 ± 10.0	1078.7 ± 5.9 *	103.6
VALO X	1310.5 ± 10.6	1172.0 ± 10.2 *	89.4	1187.0 ± 9.5	990.6 ± 12.4 *	83.5

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Bluephase PowerCure	1309.0 ± 4.9	1226.9 ± 11.6 *	93.7	1268.3 ± 1.5	1138.8 ± 10.2 *	89.8
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Blue 425 to 550 nm

LPU	Air 1 mm	FM 1 mm	T (%)	Air 2 mm	FM 2 mm	T (%)
DeepCure-S	1788.5 ± 19.3	1735.1 ± 25.5 *	97.0	1663.1 ± 26.5	1630.7 ± 8.6 *	98.0
SmartLite Pro	1168.3 ± 7.7	1163.2 ± 21.0	99.6	1035.9 ± 9.9	1073.2 ± 5.8 *	103.6
VALO X	963.1 ± 6.3	977.1 ± 9.3	101.5	887.2 ± 6.2	896.5 ± 3.9	101.0
Bluephase PowerCure	1093.2 ± 3.8	1061.8 ± 10.0 *	97.1	1061.4 ± 1.5	1018.6 ± 8.1 *	96.0

Violet 350 to 425 nm

LPU	Air 1 mm	FM 1 mm	T (%)	Air 2 mm	FM 2 mm	T (%)
DeepCure-S	28.9 ± 0.4	27.2 ± 0.4	94.1	26.9 ± 0.5	24.6 ± 0.1	91.6
SmartLite Pro	6.1 ± 0.0	6.0 ± 0.1	98.4	5.4 ± 0.1	5.5 ± 0.1	101.8
VALO X	347.4 ± 4.3	194.9 ± 5.0 *	56.1	299.8 ± 3.3	94.1 ± 14.4 *	31.4
Bluephase PowerCure	215.8 ± 1.2	165.1 ± 1.9 *	76.5	206.9 ± 0.7	120.2 ± 8.4 *	58.1

Table 3. Irradiance through EXACLEAR and through the matched air gap (mW/cm², mean ± SD, n=5). * differs from air within the same LPU (Tukey test, P<.05). T = transmission relative to air.

Total 350 to 550 nm

LPU	Air 5 mm	EXA 5 mm	T (%)	Air 10 mm	EXA 10 mm	T (%)
DeepCure-S	1264.4 ± 19.1	1317.6 ± 12.8 *	104.2	885.7 ± 6.7	901.8 ± 9.2	101.8
SmartLite Pro	830.1 ± 6.7	840.1 ± 6.1	101.2	575.7 ± 11.1	634.3 ± 5.2 *	110.2
VALO X	871.0 ± 3.1	904.3 ± 14.9	103.8	563.6 ± 3.0	595.8 ± 3.1 *	105.7
Bluephase PowerCure	926.3 ± 12.0	996.4 ± 34.1 *	107.6	417.0 ± 12.4	563.2 ± 17.1 *	135.1

Blue 425 to 550 nm

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LPU	Air 5 mm	EXA 5 mm	T (%)	Air 10 mm	EXA 10 mm	T (%)
DeepCure-S	1247.0 ± 18.8	1299.4 ± 12.6 *	104.2	874.2 ± 6.5	890.7 ± 8.9	101.9
SmartLite Pro	825.8 ± 6.6	835.9 ± 6.1	101.2	572.8 ± 11.0	631.2 ± 5.2 *	110.2
VALO X	672.5 ± 2.2	710.1 ± 10.2 *	105.6	438.1 ± 2.2	484.2 ± 2.8 *	110.5
Bluephase PowerCure	781.6 ± 9.2	854.5 ± 29.2 *	109.3	354.3 ± 11.6	489.3 ± 12.0 *	138.1

Violet 350 to 425 nm

LPU	Air 5 mm	EXA 5 mm	T (%)	Air 10 mm	EXA 10 mm	T (%)
DeepCure-S	17.4 ± 0.3	18.1 ± 0.2	104.2	11.6 ± 0.1	11.1 ± 0.3	96.2
SmartLite Pro	4.3 ± 0.1	4.2 ± 0.1	97.9	3.0 ± 0.1	3.1 ± 0.1	103.6
VALO X	198.6 ± 1.0	194.1 ± 4.9	97.8	125.4 ± 0.7	111.6 ± 1.0 *	89.0
Bluephase PowerCure	144.7 ± 3.0	141.9 ± 4.9	98.1	62.6 ± 5.2	73.9 ± 5.2 *	117.9

Table S1. Irradiance at tip contact and through the matrix materials, with transmission relative to contact (To, %) (mW/cm², mean ± SD, n=5). **Prepared at the request of Solventum (M. Agre, September 11, 2026).** To = material value divided by the contact (0 mm air) value for the same LPU; unlike Tables 2 and 3, To includes the loss attributable to tip-to-sensor distance as well as the loss attributable to the material. * differs from contact within the same LPU (Dunnett test, P<.001). One-way ANOVA (condition) was significant for every LPU and band (P<.001). Tukey test across all five conditions: every condition differed from every other within each LPU and band (P<.05), except VALO X violet, where FM 1 mm and EXA 5 mm did not differ.

Total 350 to 550 nm

LPU	Contact (0 mm)	FM 1 mm	To	FM 2 mm	To	EXA 5 mm	To	EXA 10 mm	To
DeepCure-S	1926.7 ± 13.1	1762.3 ± 25.9 *	91.5	1655.3 ± 8.6 *	85.9	1317.6 ± 12.8 *	68.4	901.8 ± 9.2 *	46.8
SmartLite Pro	1338.6 ± 39.7	1169.2 ± 21.0 *	87.3	1078.7 ± 5.9 *	80.6	840.1 ± 6.1 *	62.8	634.3 ± 5.2 *	47.4
VALO X	1507.2 ± 7.3	1172.0 ± 10.2 *	77.8	990.6 ± 12.4 *	65.7	904.3 ± 14.9 *	60.0	595.8 ± 3.1 *	39.5
Bluephase PowerCure	1368.9 ± 0.6	1226.9 ± 11.6 *	89.6	1138.8 ± 10.2 *	83.2	996.4 ± 34.1 *	72.8	563.2 ± 17.1 *	41.1

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Blue 425 to 550 nm

LPU	Contact (0 mm)	FM 1 mm	T ₀	FM 2 mm	T ₀	EXA 5 mm	T ₀	EXA 10 mm	T ₀
DeepCure-S	1896.2 ± 12.9	1735.1 ± 25.5 *	91.5	1630.7 ± 8.6 *	86.0	1299.4 ± 12.6 *	68.5	890.7 ± 8.9 *	47.0
SmartLite Pro	1331.6 ± 39.4	1163.2 ± 21.0 *	87.4	1073.2 ± 5.8 *	80.6	835.9 ± 6.1 *	62.8	631.2 ± 5.2 *	47.4
VALO X	1084.0 ± 4.7	977.1 ± 9.3 *	90.1	896.5 ± 3.9 *	82.7	710.1 ± 10.2 *	65.5	484.2 ± 2.8 *	44.7
Bluephase PowerCure	1141.2 ± 0.5	1061.8 ± 10.0 *	93.0	1018.6 ± 8.1 *	89.3	854.5 ± 29.2 *	74.9	489.3 ± 12.0 *	42.9

Violet 350 to 425 nm

LPU	Contact (0 mm)	FM 1 mm	T ₀	FM 2 mm	T ₀	EXA 5 mm	T ₀	EXA 10 mm	T ₀
DeepCure-S	30.5 ± 0.3	27.2 ± 0.4 *	89.2	24.6 ± 0.1 *	80.7	18.1 ± 0.2 *	59.3	11.1 ± 0.3 *	36.4
SmartLite Pro	7.0 ± 0.3	6.0 ± 0.1 *	85.7	5.5 ± 0.1 *	78.6	4.2 ± 0.1 *	60.0	3.1 ± 0.1 *	44.3
VALO X	423.2 ± 2.6	194.9 ± 5.0 *	46.1	94.1 ± 14.4 *	22.2	194.1 ± 4.9 *	45.9	111.6 ± 1.0 *	26.4
Bluephase PowerCure	227.7 ± 0.2	165.1 ± 1.9 *	72.5	120.2 ± 8.4 *	52.8	141.9 ± 4.9 *	62.3	73.9 ± 5.2 *	32.5

Findings, Table S1. Relative to tip contact, every material condition delivered significantly less irradiance for every LPU and band. Blue light through Filtek Matrix reached 91 to 93% of contact at 1 mm and 81 to 89% at 2 mm for all four LPUs. Violet light through Filtek Matrix for the multi-peak LPUs reached 46% (VALO X) and 73% (PowerCure) of contact at 1 mm and 22% and 53% at 2 mm. Through EXACLEAR, blue light reached 63 to 75% of contact at 5 mm and 43 to 47% at 10 mm, values that are equal to or above the corresponding air gap alone (Table 1), consistent with the light-guiding effect described in Section 6.

Table 4. Bulk attenuation coefficients (% per mm), Beer-Lambert calculation between the two thicknesses of each material. The interface reflection term cancels. Section 1A coefficients for EXACLEAR are dominated by light guiding and are not reported; single-peak units emit too little violet for a coefficient.

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LPU	Section 1B, Filtek Matrix (all light)	Section 1B, EXACLEAR (all light)	Section 1A, Filtek Matrix, blue	Section 1A, Filtek Matrix, violet
DeepCure-S	3.6	1.7	within noise	not estimable
SmartLite Pro	2.4	0.9	within noise	not estimable
VALO X	3.5	1.5	0.4	58
Bluephase PowerCure	2.2	1.2	1.2	28

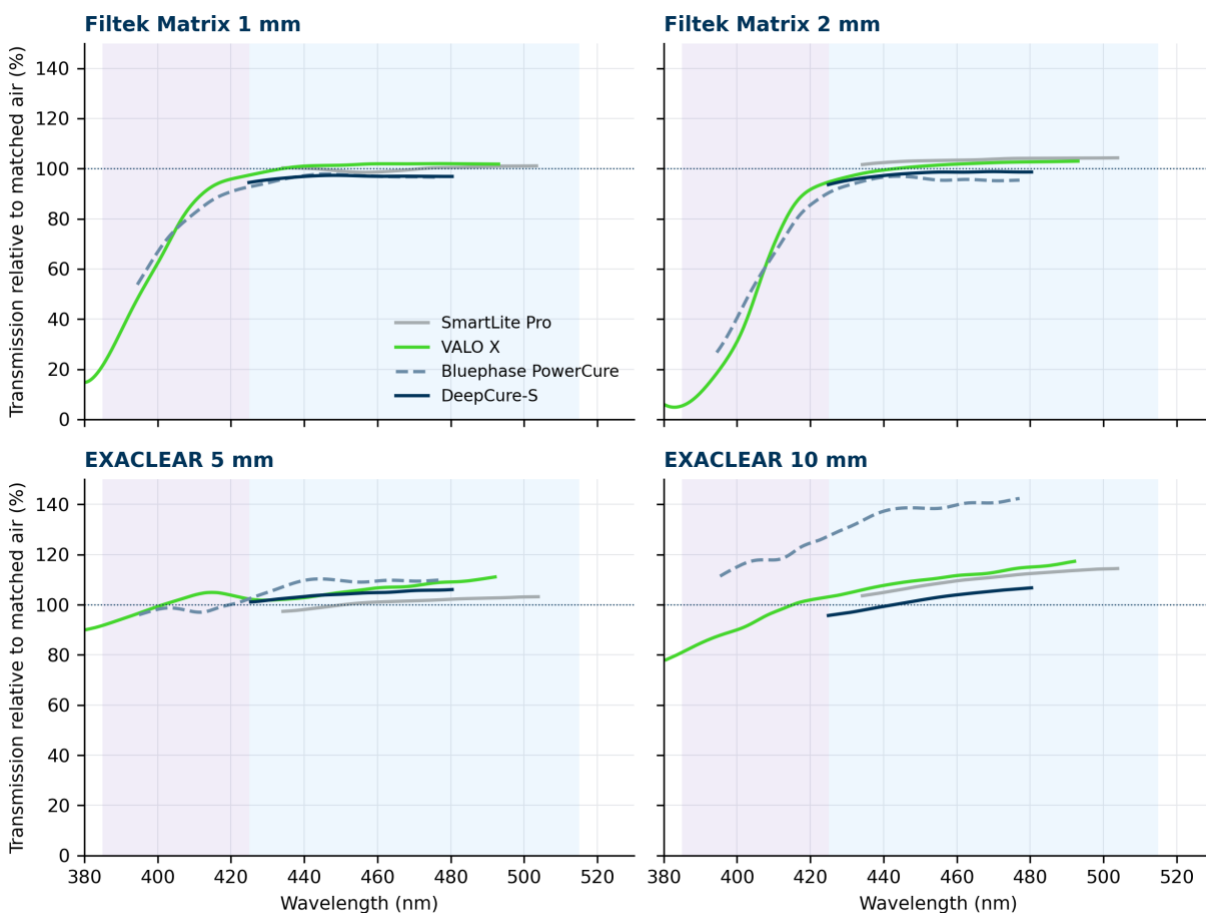


Figure 4. Transmission relative to the matched air gap as a function of wavelength, computed as the mean through-material spectrum divided by the mean air spectrum at the same distance (5 nm moving average) for each LPU and material. Curves are drawn only where the air spectrum exceeds 5% of that unit's peak, so single-peak units appear only above 425 nm. Shaded bands mark the violet (385 to 425 nm) and blue (425 to 515 nm) ranges.

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Findings, 1A. Ten millimeters of air reduced irradiance to 30 to 46% of the contact value for every unit. Filtek Matrix transmitted 96 to 104% of blue light for all four units at both thicknesses, but only 31 to 56% (VALO X) and 58 to 77% (PowerCure) of violet light; bulk violet attenuation of 28 to 58% per mm identifies the material as a violet absorber. Figure 4 resolves the cutoff: through 2 mm of Filtek Matrix, transmission is 95% at 425 nm, 85% at 415 nm, 50% at 405 nm, 19% at 395 nm, and 5% at 385 nm, and VALO X and PowerCure fall on the same curve, so the loss is a property of the material rather than the unit. EXACLEAR shows no wavelength cutoff, only a gain that increases with wavelength and thickness. EXACLEAR delivered 101 to 138% of the matched air value in the total and blue bands because the block guides into the sensor light that the bare beam spreads past.

4.2 Section 1B. Beam profile

Method. The beam profiler (3.45 μm sensor pitch, 3.766 $\times$ optical scaling, 12.99 μm effective pixel) was power calibrated at contact for each LPU against the radiant power measured with the integrating sphere spectroradiometer on May 20, 2026 (Table M1). Conditions were as in 1A, but material specimens were placed in the air path between tip and target. One calibrated frame per condition (36 frames) was extracted from the native .bgData files, pedestal corrected, and analyzed for power within a fixed 2.544 cm^2 aperture, D4 σ diameter, mean irradiance within the D4 σ circle, peak irradiance, and plateau coefficient of variation. In plain terms, D4 σ treats the beam like a pile of sand: it measures how widely the light is spread around the center, using every pixel weighted by its brightness. It needs no arbitrary cutoff for where the beam ends, but it is sensitive to faint light at the margins, so a small amount of light scattered outside the edge makes the beam read slightly wider even if its bright core has not changed. For flat beams such as VALO X, D4 σ is smaller than the edge-to-edge diameter seen in Figure 1, so it is used here to compare conditions rather than to describe the size of the tip footprint. Plateau coefficient of variation describes how even the light is across the bright part of the beam; lower values mean a flatter, more uniform beam. Frames were centroid registered and material-minus-air difference maps were generated at matched distances. No inferential statistics were applied.

Deviation from the proposal. A single 8 mm distance with five replicates was proposed. The full 0 to 10 mm series was captured with one frame each to allow matched-distance comparison with 1A.

Table 5. Power within the 2.544 cm^2 aperture (mW)

LPU	0 mm	1 mm	FM 1 mm	2 mm	FM 2 mm	5 mm	EXA 5 mm	10 mm	EXA 10 mm
DeepCure-S	711	706	639	703	613	690	602	668	537

SmartLite Pro	983	979	900	977	877	954	849	910	775
VALO X	1306	1302	1175	1293	1127	1267	1095	1223	982
Bluephase PowerCure	566	563	512	559	497	544	497	517	446

Table 6. Mean irradiance within the D4σ beam diameter (mW/cm²)

LPU	0 mm	1 mm	FM 1 mm	2 mm	FM 2 mm	5 mm	EXA 5 mm	10 mm	EXA 10 mm
DeepCure-S	1129	1123	1006	1109	953	1025	891	838	708
SmartLite Pro	1178	1151	1003	1120	953	985	857	774	677
VALO X	1022	1012	899	993	848	922	796	793	656
Bluephase PowerCure	1071	1061	926	1028	878	872	800	597	565

Table 7. Peak irradiance (mW/cm²)

LPU	0 mm	1 mm	FM 1 mm	2 mm	FM 2 mm	5 mm	EXA 5 mm	10 mm	EXA 10 mm
DeepCure-S	1815	1573	1465	1519	1350	1482	1293	1472	1156
SmartLite Pro	1843	1793	1612	1759	1451	1606	1497	1511	1293
VALO X	1649	1709	1468	1677	1286	1640	1231	1561	1163
Bluephase PowerCure	1572	1491	1352	1471	1316	1441	1318	1285	1138

SmartLite Pro values in Tables 5 to 7 are rescaled by 983/884 (Appendix C).

Table 8. D4σ beam diameter (cm)

LPU	0 mm	1 mm	FM 1 mm	2 mm	FM 2 mm	5 mm	EXA 5 mm	10 mm	EXA 10 mm
DeepCure-S	0.876	0.875	0.876	0.875	0.879	0.891	0.893	0.953	0.932
SmartLite Pro	1.011	1.015	1.032	1.022	1.041	1.062	1.072	1.153	1.140
VALO X	1.252	1.255	1.260	1.259	1.266	1.281	1.281	1.336	1.322
Bluephase PowerCure	0.801	0.802	0.811	0.806	0.817	0.848	0.846	0.983	0.942

Table 9. Plateau coefficient of variation (pixels above 50% of peak; lower = flatter)

LPU	0 mm	1 mm	FM 1 mm	2 mm	FM 2 mm	5 mm	EXA 5 mm	10 mm	EXA 10 mm
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DeepCure-S	0.151	0.124	0.122	0.128	0.126	0.144	0.142	0.154	0.157
SmartLite Pro	0.106	0.105	0.111	0.108	0.125	0.129	0.121	0.133	0.133
VALO X	0.110	0.100	0.099	0.094	0.103	0.078	0.093	0.070	0.080
Bluephase PowerCure	0.147	0.144	0.158	0.155	0.164	0.180	0.186	0.185	0.185

Table 10. Transmission relative to the matched air gap, power within aperture (%). Loss model with the four LPUs as replicates (n=8 per material): Filtek Matrix 6.4% + 2.6%/mm (R^2 0.67); EXACLEAR 5.9% + 1.1%/mm (R^2 0.58). The intercepts are consistent with Fresnel reflection at two interfaces of refractive index 1.4 to 1.5.

LPU	FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm
DeepCure-S	90.5	87.3	87.2	80.3
SmartLite Pro	92.0	89.8	89.1	85.1
VALO X	90.3	87.1	86.4	80.2
Bluephase PowerCure	90.9	89.0	91.4	86.3

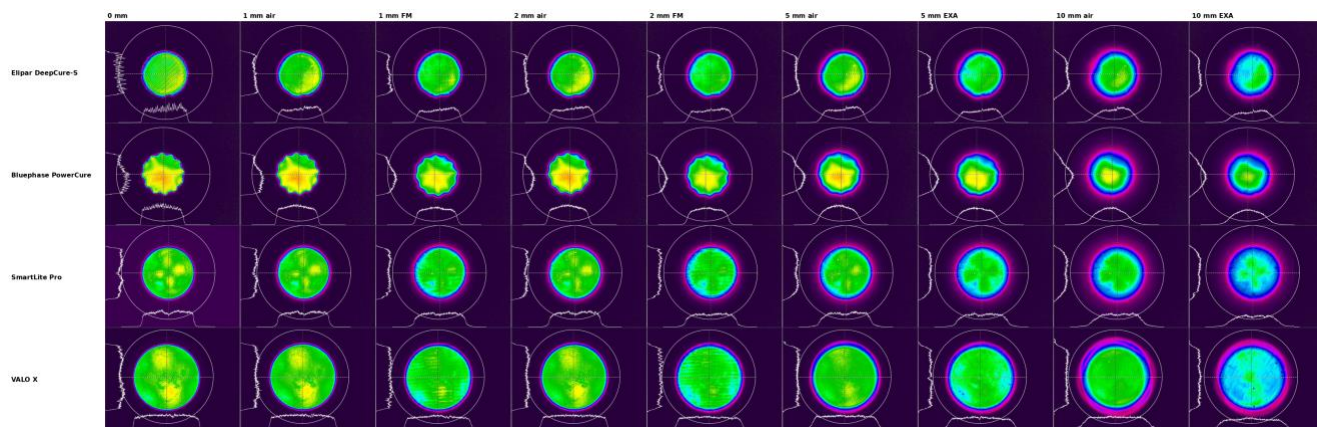


Figure S1. Screenshots of the BeamGage display for the same 36 conditions shown in Figure 1. Rows: Elipar DeepCure-S, Bluephase PowerCure, SmartLite Pro, VALO X. Columns: tip in contact (0 mm); 1 mm air; 1 mm Filtek Matrix (FM); 2 mm air; 2 mm FM; 5 mm air; 5 mm EXACLEAR (EXA); 10 mm air; 10 mm EXA. For each unit, exposure was set on the contact frame to just below saturation and then held fixed for every interposed condition, so panels are comparable down a row but not between rows. Figure 1 is the calibrated version of the same data on one common scale.

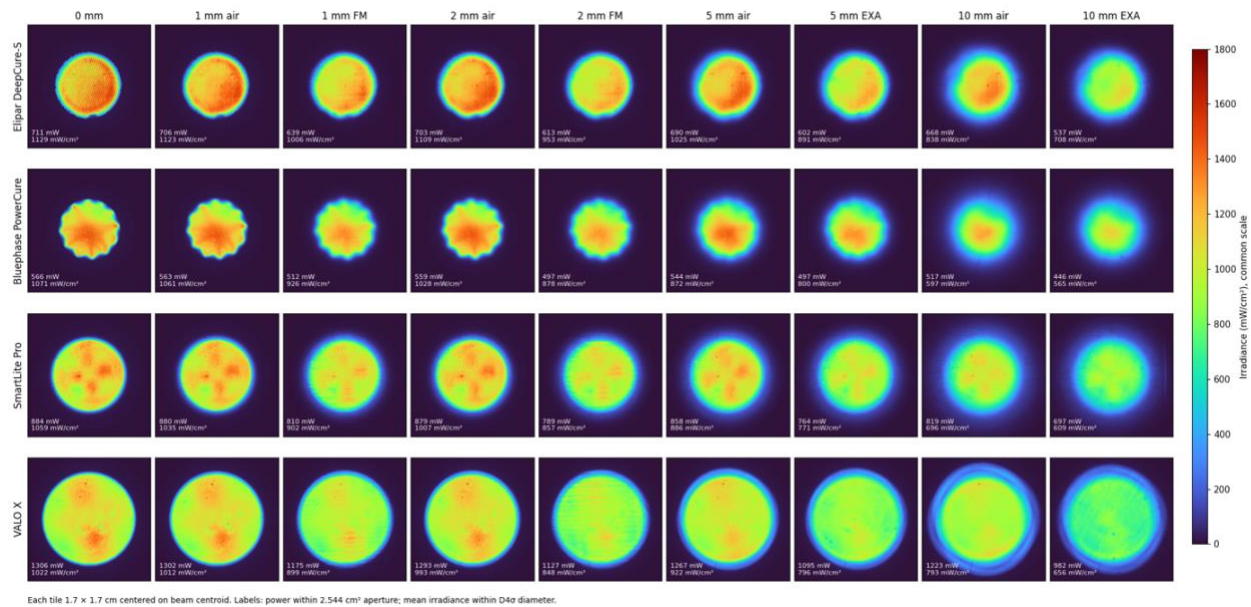


Figure 1. Calibrated irradiance maps of four LPUs in air and through light-transmitting matrix materials, displayed on a common scale (0 to 1800 mW/cm²). Rows: Elipar DeepCure-S, Bluephase PowerCure, SmartLite Pro, VALO X. Columns: tip in contact (0 mm); 1 mm air; 1 mm Filtek Matrix (FM); 2 mm air; 2 mm FM; 5 mm air; 5 mm EXACLEAR (EXA); 10 mm air; 10 mm EXA. Each tile is 1.7 × 1.7 cm centered on the beam centroid. Labels give power within the 2.544 cm² aperture and mean irradiance within the D4σ diameter.

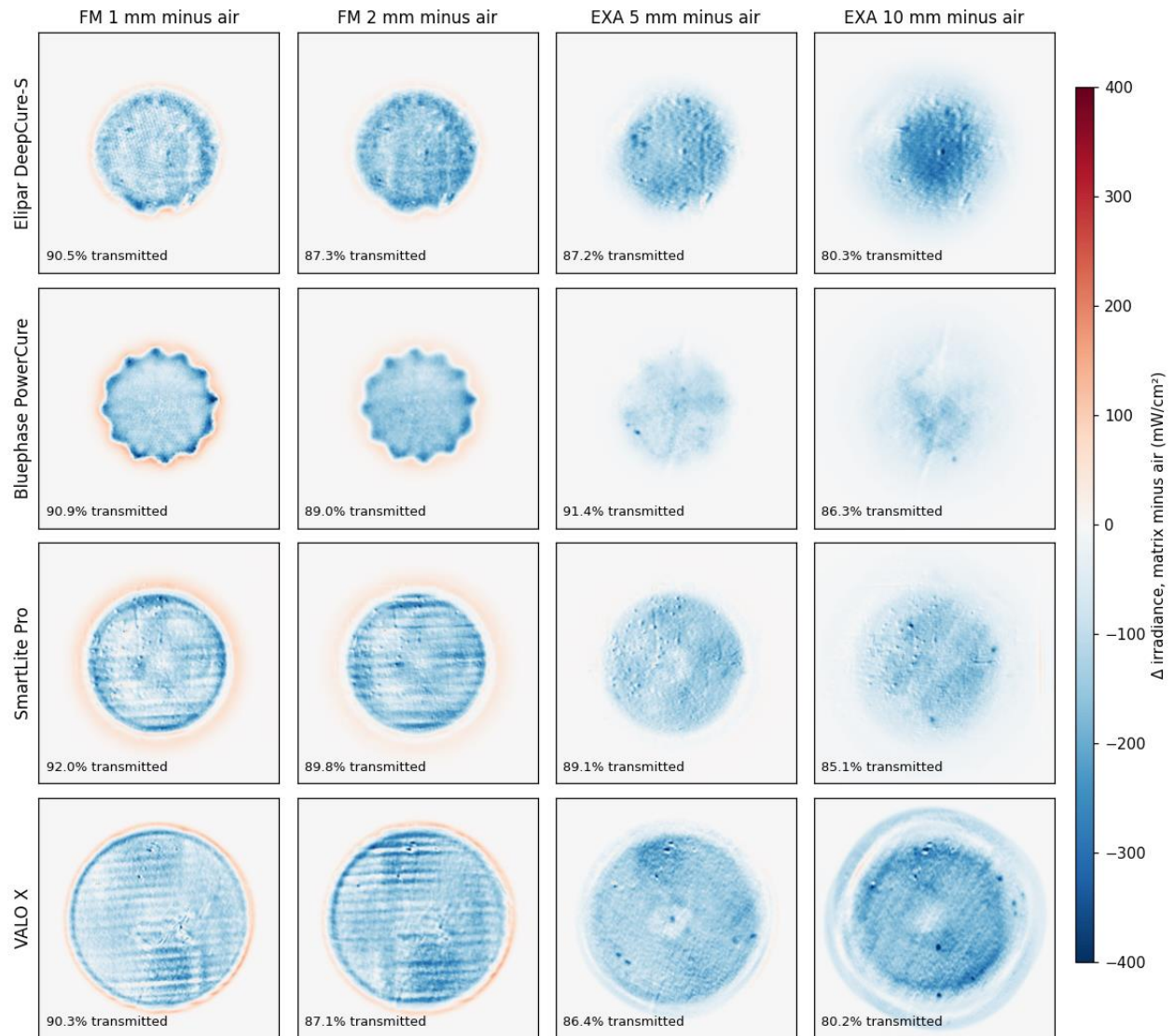


Figure 2. Difference maps, material minus air at matched distance, after centroid registration (scale ± 400 mW/cm²; blue indicates loss through the material, red indicates gain). Rows as in Figure 1. Columns: FM 1 mm minus 1 mm air; FM 2 mm minus 2 mm air; EXA 5 mm minus 5 mm air; EXA 10 mm minus 10 mm air. Labels give transmission of power within the aperture relative to the matched air gap.

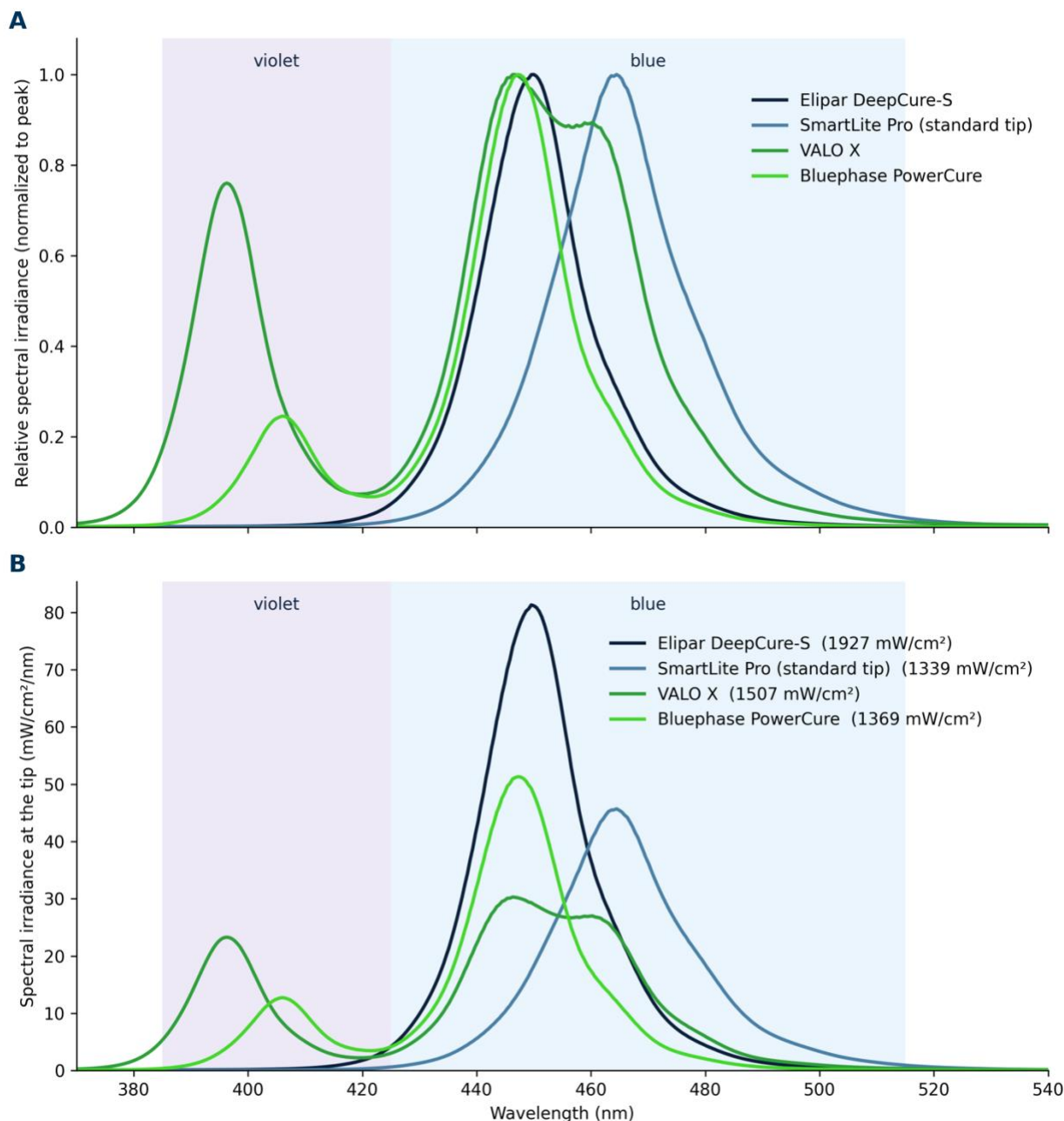


Figure 3. Emission spectra of the four LPUs at contact, measured with the cosine-corrector spectroradiometer (Section 1A, 0 mm). A, normalized to each unit's peak. B, absolute spectral irradiance at the tip, with the integrated 350 to 550 nm irradiance of Table 1 in the legend. Shaded bands mark the violet (385 to 425 nm) and blue (425 to 515 nm) ranges.

Findings, 1B. Figure 1 shows that each unit has a distinct footprint at contact: a fiber-rod mosaic in a round outline with a flattened segment at the bottom border (DeepCure-S), a scalloped

lens outline with a central peak (PowerCure), a mottled disk with three or four discrete hot regions (SmartLite Pro), and a large flat disk with a sharp rim and a hot region below center (VALO X). These signatures survive both materials in every column. Through Filtek Matrix, each beam is a slightly softened version of its air counterpart: edges blur, the DeepCure-S mosaic washes out by 1 mm, hot-spot positions do not move, and a fine horizontal striation appears on SmartLite Pro and VALO X that is absent in the matched air columns and is consistent with the printed surface of the matrix specimen. Figure 2 shows the mechanism: through Filtek Matrix every unit loses 100 to 200 mW/cm² uniformly across the plateau and gains a thin red ring just outside the beam edge, meaning the material redistributes light outward rather than absorbing selectively; on PowerCure the scallop tips alternate blue and red as the matrix blurs the lens pattern. Through EXACLEAR at 10 mm the loss concentrates toward the center on DeepCure-S and VALO X, and VALO X loses the concentric outer ring it shows in air (Figure 1, column 8). PowerCure collapses from a top-hat to a near-Gaussian profile by 5 mm in air (Figure 1, columns 6 and 8; CV 0.147 to 0.185), and neither material alters that collapse. D4σ through Filtek Matrix is 0 to 2% larger than in air at the same distance and through EXACLEAR up to 1% larger to 4% smaller, and plateau CV changes by no more than 0.02 (Tables 8 and 9). Figure 3 shows why the violet finding of Section 1A applies to two of the four units: VALO X and PowerCure emit 25% and 17% of their output below 425 nm, DeepCure-S and SmartLite Pro emit 1.2% and 0.2%. In absolute terms (Figure 3B), the VALO X violet peak reaches 23 mW/cm²/nm and the PowerCure peak 13 mW/cm²/nm, against blue peaks of 81 (DeepCure-S), 51 (PowerCure), 46 (SmartLite Pro), and 30 (VALO X) mW/cm²/nm; the unit with the most violet also has the lowest blue spectral irradiance. SmartLite Pro peaks at 464 nm, further from the camphorquinone absorption maximum than the other three units, which peak at 446 to 449 nm.

Limitation. The camera and target are not spectrally flat and the power calibration was set in air. Through a violet-attenuating material the calibration factor for multi-peak units is no longer exact, which likely explains why the profiler shows a uniform 10% loss where the spectroradiometer separates the units.

4.3 Section 1C. Reconciliation of the two methods

Table 11. Transmission relative to matched air, 1A (spectroradiometer, material coupled to the sensor) and 1B (beam profiler, material in the air path), %

Material	LPU	1A total 350 to 550 nm	1A blue 425 to 550 nm	1B power in aperture	1B central 4 mm	1A total minus 1B aperture
	DeepCure-S	97.0	97.0	90.5	91	+6.5

Filtek Matrix 1 mm	SmartLite Pro	99.6	99.6	92.0	91	+7.6
	VALO X	89.4	101.5	90.3	91	-0.9
	Bluephase PowerCure	93.7	97.1	90.9	92	+2.8
Filtek Matrix 2 mm	DeepCure-S	97.9	98.0	87.3	88	+10.6
	SmartLite Pro	103.6	103.6	89.8	89	+13.8
	VALO X	83.5	101.0	87.1	89	-3.6
	Bluephase PowerCure	89.8	96.0	89.0	89	+0.8
EXACLEAR 5 mm	DeepCure-S	104.2	104.2	87.2	87	+17.0
	SmartLite Pro	101.2	101.2	89.1	90	+12.1
	VALO X	103.8	105.6	86.4	89	+17.4
	Bluephase PowerCure	107.6	109.3	91.4	92	+16.2
EXACLEAR 10 mm	DeepCure-S	101.8	101.9	80.3	79	+21.5
	SmartLite Pro	110.2	110.2	85.1	87	+25.1
	VALO X	105.7	110.5	80.2	84	+25.5
	Bluephase PowerCure	135.1	138.1	86.3	89	+48.8

Findings, 1C. For Filtek Matrix the two methods agree within 4 points on the multi-peak units and differ by 7 to 14 points on the single-peak units. For EXACLEAR they differ in sign. Both rank the units identically and agree on the direction of the Filtek Matrix effect. The reasons are given in Section 6.

5. Part 2. Effect on resin composite polymerization

Two LPUs were carried into Part 2: Elipar DeepCure-S (single-peak, highest blue irradiance) and VALO X (multi-peak, largest violet loss through the matrix). Two resin composites were used, both shade A2 (Table M2): Filtek Supreme Ultra (camphorquinone system) and Vit-l-escence (camphorquinone with an alternative photoinitiator). Specimens were fabricated and measured in randomized order. Five light paths were tested: tip in contact (air), Filtek Matrix 1 and 2 mm, and EXACLEAR 5 and 10 mm, with the LPU tip placed directly on the material specimen. Exposure was 20 seconds. Radiant exposure by condition, from Table 6, ranged from 13.1 to 22.6 J/cm².

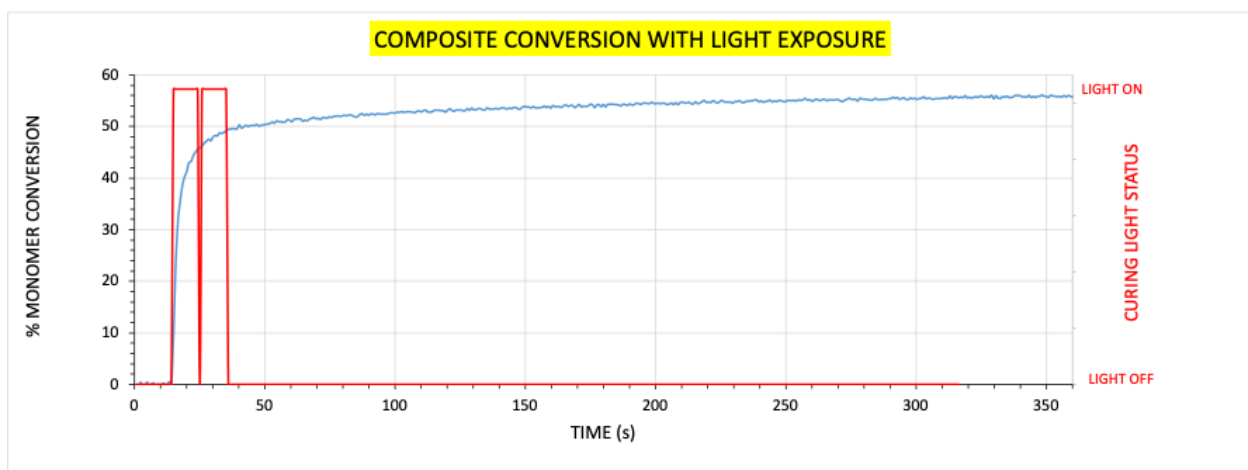
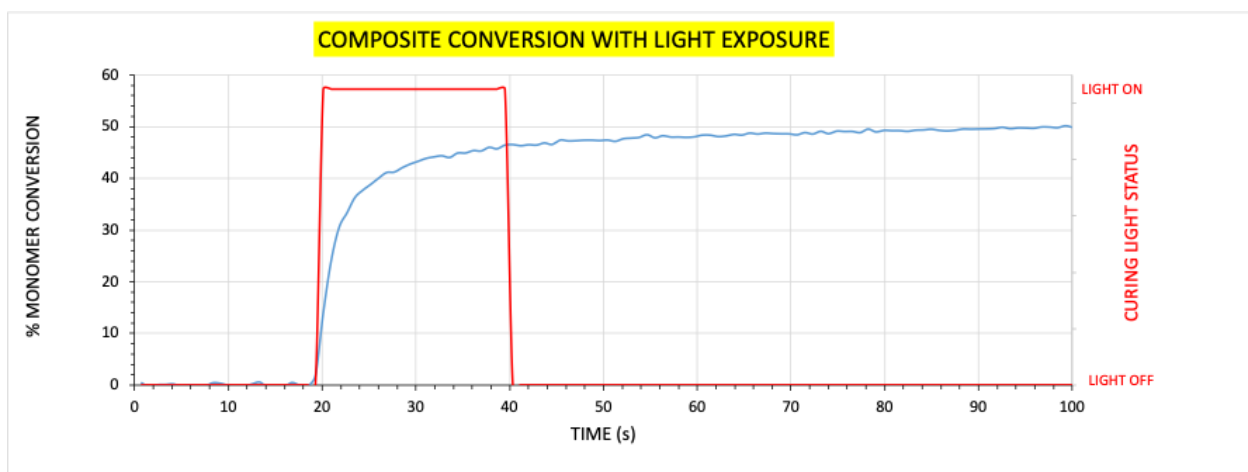
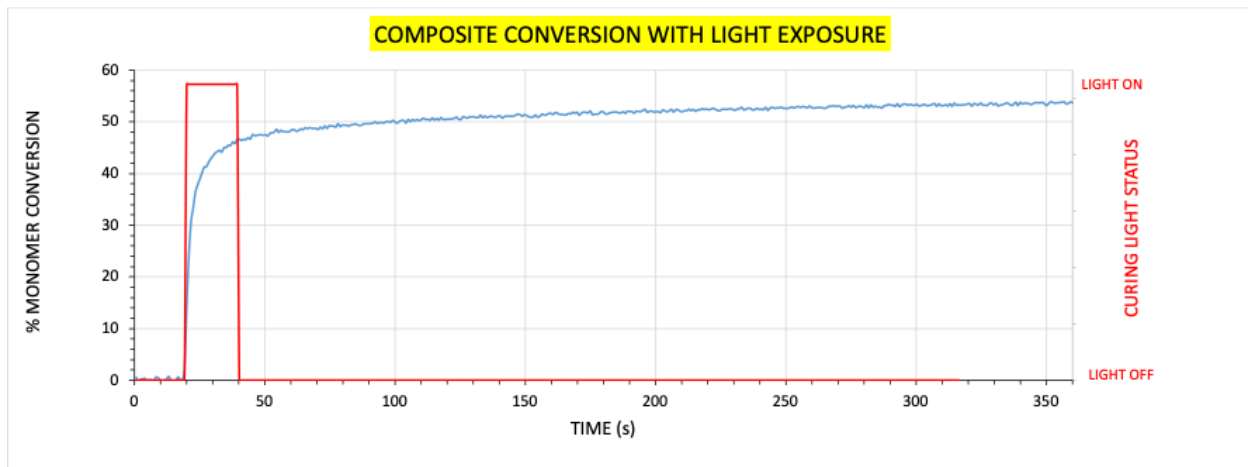
5.1 Section 2A. Degree of conversion

Method. Thin films of each resin composite were placed on the ATR crystal of the FTIR spectrometer, the material specimen was placed over the film, and the LPU tip was placed on the specimen. Spectra were acquired continuously for 6 minutes in kinetic mode, and light activation was recorded on an external photodetector channel on the same time base. DC was computed from the 1636 cm^{-1} (aliphatic) to 1608 cm^{-1} (aromatic) C=C peak-height ratio (OPUS integration Method Q; peak search windows 1640 to 1631 and 1614 to 1604 cm^{-1} ; sloped baseline anchored at 1656 to 1654 and 1591 to 1589 cm^{-1}), referenced to the ratio immediately before light activation. Peak-height ratios were converted to conversion with the quadratic calibration of Rueggeberg et al. ($1.029R + 0.173R^2$) to correct for the nonlinearity of the aliphatic-to-aromatic ratio; the uncured reference ratio was the mean of the first ten spectra before light activation. The reported DC is the value 300 seconds after light initiation, one value per specimen, $n=5$ per condition. Three-way ANOVA (LPU $\times$ resin composite $\times$ condition) was performed with the Tukey test and Dunnett comparisons against air ($\alpha=.05$).

Deviation from the proposal. DC at the bottom of 2 mm specimens was proposed. Thin-film DC was measured instead because the shrinkage of the resin composite was causing it to debond from the ATR crystal. The proposal also specified three readings averaged per specimen; the kinetic acquisition yields one continuous trace per specimen, from which the single 300-second value was taken.

Table 12. Thin-film degree of conversion 300 seconds after light initiation (%), mean $\pm$ SD, $n=5$. No condition differed from air within any LPU and resin composite pairing (Dunnett test, all $P \geq .065$). Uppercase letters: LPU and resin composite pairings pooled across conditions (Tukey test, $\alpha=.05$).

LPU / resin composite	Air (contact)	FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm	Pooled
DeepCure-S / Filtek Supreme Ultra	54.0 ± 0.8	53.9 ± 1.0	53.2 ± 0.4	53.3 ± 0.4	53.3 ± 0.4	53.5 ± 0.7 C
DeepCure-S / Vit-l- escence	53.7 ± 0.7	54.1 ± 1.0	54.8 ± 1.1	54.4 ± 0.3	53.7 ± 0.2	54.2 ± 0.8 B
VALO X / Filtek Supreme Ultra	54.0 ± 0.2	53.3 ± 1.1	52.8 ± 1.2	52.8 ± 1.3	52.5 ± 0.2	53.1 ± 1.0 C
VALO X / Vit-l-escence	55.9 ± 0.5	55.5 ± 0.3	55.8 ± 0.4	55.5 ± 0.9	55.3 ± 0.3	55.6 ± 0.5 A



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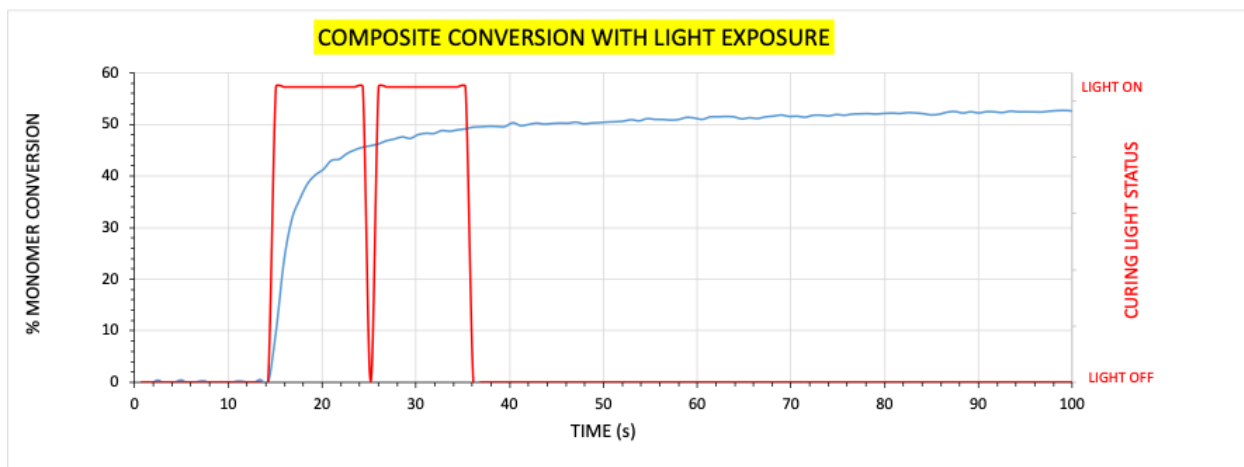


Figure S2. Representative real-time conversion traces. Blue, monomer conversion of the resin composite; red, curing light status recorded on the external photodetector channel of the spectrometer on the same time base. A and B, a unit delivering a single continuous 20 second exposure, shown over the full 6 minute acquisition (A) and over the first 100 seconds (B). C and D, VALO X, whose maximum continuous exposure is 10 seconds, so the 20 seconds were delivered as two consecutive 10 second exposures, again shown full length (C) and over the first 100 seconds (D). Conversion rises while the light is on and continues to climb slowly in the dark. The values in Table 12 are read from these traces at 300 seconds after light initiation. A trace of this kind exists for every degree of conversion specimen in the study and all are available on request.

Findings, 2A. No matrix or index condition changed DC for any unit and resin composite pairing. The condition main effect ($P=.043$) and condition $\times$ resin composite interaction ($P=.019$) reflect a 1 point drift within Filtek Supreme Ultra that no pairwise comparison isolates. The substantive effect is the LPU $\times$ resin composite interaction ($P<.001$): VALO X converted Vit-I-escence about 1.5 points higher than DeepCure-S at every condition, including through 2 mm of matrix, whereas Filtek Supreme Ultra converted equally with either unit.

5.2 Section 2B. Knoop hardness and bottom-to-top ratio

Method. Disks 5.0 mm in diameter and 2.0 mm thick were packed into 3D-printed tooth-shaded resin molds on a glass slide over white paper, covered with Mylar, pressed flat with a second slide, the slide was removed, the material specimen was placed on the Mylar, and the LPU tip was placed on the specimen. Specimens were stored dry and dark at 37 °C for 24 hours. Five Knoop indentations per surface were made in a fixed five-point pattern (50 gf load, 15-second dwell) on top and bottom. Outcomes were top KHN, bottom KHN, and the bottom-to-top (B/T) ratio. Three-way ANOVA was performed with the Tukey test and Dunnett comparisons against air ($\alpha=.05$). Unmeasurable indentations on three DeepCure-S / Filtek Supreme Ultra specimens were

excluded before analysis: top and bottom surfaces of two FM 2 mm specimens and the bottom surface of one EXACLEAR 5 mm specimen (indentation corners not resolvable because of surface artifacts). The analysis with these values retained is in Appendix B and does not change any conclusion.

Deviations from the proposal. Knoop instead of Vickers; five indents per surface instead of fifteen; 5 mm disks instead of 10 mm; top surface added. With beam diameters of 0.8 to 1.3 cm, the 5 mm disk sits within the plateau of every unit, which removes edge-of-beam variation but means the data cannot map spatial nonuniformity.

Table 13. Top-surface Knoop hardness at 24 hours (KHN), mean $\pm$ SD (n=5; n=3 where noted). No condition differed from air within any pairing (Dunnett test, $P>.05$). Uppercase letters: pairings pooled across conditions (Tukey test, $\alpha=.05$).

LPU / resin composite	Air (contact)	FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm	Pooled
DeepCure-S / Filtek Supreme Ultra	73.6 $\pm$ 1.9	72.9 $\pm$ 6.4	72.8 $\pm$ 3.8 (n=3)	73.8 $\pm$ 3.4	75.5 $\pm$ 4.4	73.8 $\pm$ 4.0 A
DeepCure-S / Vit-I-escence	55.8 $\pm$ 4.9	52.9 $\pm$ 7.2	56.5 $\pm$ 3.4	51.6 $\pm$ 10.3	52.2 $\pm$ 5.7	53.8 $\pm$ 6.5 C
VALO X / Filtek Supreme Ultra	73.8 $\pm$ 3.0	67.2 $\pm$ 9.1	67.5 $\pm$ 6.5	74.9 $\pm$ 4.1	71.8 $\pm$ 3.6	71.0 $\pm$ 6.1 A
VALO X / Vit-I-escence	57.6 $\pm$ 5.2	59.3 $\pm$ 5.7	58.3 $\pm$ 2.5	61.7 $\pm$ 10.8	54.9 $\pm$ 4.0	58.4 $\pm$ 6.2 B

Table 14. Bottom-surface Knoop hardness at 24 hours (KHN), mean $\pm$ SD (n=5; n=3 to 4 where noted). No condition differed from air within any pairing (Dunnett test, $P>.05$). Uppercase letters: pairings pooled across conditions (Tukey test, $\alpha=.05$).

LPU / resin composite	Air (contact)	FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm	Pooled
DeepCure-S / Filtek Supreme Ultra	71.9 $\pm$ 2.0	71.4 $\pm$ 5.4	68.0 $\pm$ 1.5 (n=3)	69.0 $\pm$ 4.2 (n=4)	76.3 $\pm$ 3.4	71.7 $\pm$ 4.4 A
DeepCure-S / Vit-I-escence	42.5 $\pm$ 7.2	43.7 $\pm$ 4.4	47.3 $\pm$ 3.0	42.5 $\pm$ 14.7	41.9 $\pm$ 6.5	43.6 $\pm$ 7.7 B
VALO X / Filtek Supreme Ultra	69.0 $\pm$ 6.5	62.3 $\pm$ 5.4	66.0 $\pm$ 6.9	66.4 $\pm$ 7.3	68.3 $\pm$ 3.4	66.4 $\pm$ 6.0 A
VALO X / Vit-I-escence	36.7 $\pm$ 11.2	36.0 $\pm$ 9.7	36.6 $\pm$ 6.8	41.0 $\pm$ 12.6	36.4 $\pm$ 5.1	37.3 $\pm$ 8.9 C

Table 15. Bottom-to-top hardness ratio, mean $\pm$ SD (n=5; n=3 to 4 where noted). No condition differed from air within any pairing (Dunnett test, $P>.05$). Uppercase letters: pairings pooled across conditions (Tukey test, $\alpha=.05$).

LPU / resin composite	Air (contact)	FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm	Pooled
DeepCure-S / Filtek Supreme Ultra	0.98 $\pm$ 0.03	0.98 $\pm$ 0.03	0.94 $\pm$ 0.06 (n=3)	0.93 $\pm$ 0.08 (n=4)	1.01 $\pm$ 0.03	0.97 $\pm$ 0.05 A
DeepCure-S / Vit-I-escence	0.76 $\pm$ 0.08	0.84 $\pm$ 0.17	0.84 $\pm$ 0.04	0.81 $\pm$ 0.19	0.80 $\pm$ 0.09	0.81 $\pm$ 0.12 B
VALO X / Filtek Supreme Ultra	0.94 $\pm$ 0.10	0.93 $\pm$ 0.09	0.98 $\pm$ 0.11	0.89 $\pm$ 0.08	0.95 $\pm$ 0.02	0.94 $\pm$ 0.08 A
VALO X / Vit-I-escence	0.63 $\pm$ 0.14	0.61 $\pm$ 0.18	0.63 $\pm$ 0.11	0.65 $\pm$ 0.12	0.67 $\pm$ 0.09	0.64 $\pm$ 0.12 C

Table 16. Three-way ANOVA summary, Part 2 (P value; partial η^2 in parentheses)

Source	DC	Top KHN	Bottom KHN	B/T ratio
Condition	.043 (0.11)	.636 (0.03)	.887 (0.01)	.769 (0.02)
LPU	.001 (0.13)	.356 (0.01)	<.001 (0.16)	<.001 (0.24)
Composite	<.001 (0.59)	<.001 (0.71)	<.001 (0.83)	<.001 (0.61)
Condition $\times$ LPU	.361 (0.05)	.374 (0.05)	.684 (0.03)	.936 (0.01)
Condition $\times$ resin composite	.019 (0.14)	.314 (0.06)	.410 (0.05)	.748 (0.02)
LPU $\times$ resin composite	<.001 (0.35)	.003 (0.11)	.709 (0.00)	.002 (0.12)
Three-way	.753 (0.02)	.706 (0.03)	.776 (0.02)	.683 (0.03)

Findings, 2B. Light path did not affect top KHN, bottom KHN, or B/T ratio, did not interact with unit or resin composite, and no condition differed from air in any pairing. Composite explained 61 to 83% of variance. The LPU $\times$ resin composite interaction for top KHN ($P=.003$) and B/T ($P=.002$) shows that **VALO X made Vit-I-escence harder on top and softer at the bottom than DeepCure-S did** (B/T 0.61 to 0.67 compared with 0.76 to 0.84), **whereas Filtek Supreme Ultra was unaffected by unit choice**. Vit-I-escence with VALO X falls below the 0.80 B/T criterion even in direct contact; the matrix adds no further penalty, but this pairing does not reach adequate depth of polymerization at 2 mm under any condition tested.

5.3 Section 2C. Correlation of delivered light with polymerization outcomes

Twenty groups were analyzed (2 LPU's × 2 resin composites × 5 conditions). Delivered light was taken from Table 6 (1B mean irradiance within D4σ, complete for all 20) and from Tables 1 to 3 (1A blue and violet irradiance at the matched condition).

Table 17. Pearson r (P) across all 20 groups

Delivered light	DC	Top KHN	Bottom KHN	B/T ratio
1B irradiance within D4σ	0.13 (.586)	0.01 (.971)	0.04 (.869)	0.07 (.764)
1A blue irradiance	-0.05 (.838)	-0.03 (.900)	0.12 (.608)	0.26 (.276)
1A violet irradiance	0.33 (.161)	0.09 (.707)	-0.13 (.592)	-0.31 (.188)

Table 18. Pearson r (P) within each resin composite

Composite	Delivered light	DC	Bottom KHN	B/T ratio
Filtek Supreme Ultra	1B irradiance	0.83 (.003)	0.01 (.980)	0.01 (.975)
	1A blue	0.75 (.012)	0.30 (.403)	0.27 (.451)
	1A violet	0.14 (.697)	-0.41 (.242)	-0.46 (.182)
Vit-l-escence	1B irradiance	-0.19 (.604)	0.31 (.388)	0.21 (.565)
	1A blue	-0.59 (.072)	0.69 (.028)	0.66 (.038)
	1A violet	0.75 (.013)	-0.62 (.054)	-0.74 (.014)

Table 19. DC compared with hardness

Comparison	r (P)
DC and top KHN, all 20 groups	-0.59 (.006)
DC and bottom KHN, all 20 groups	-0.78 (<.001)
DC and B/T ratio, all 20 groups	-0.85 (<.001)
DC and bottom KHN, Filtek Supreme Ultra only	0.41 (.241)
DC and bottom KHN, Vit-l-escence only	-0.66 (.036)
DC and top KHN, Vit-l-escence only	0.73 (.016)

Findings, 2C. Across all 20 groups, delivered irradiance predicts no outcome. Within Filtek Supreme Ultra, thin-film DC tracks delivered irradiance (r 0.75 to 0.84) whereas hardness does not. Within Vit-l-escence, violet irradiance raises surface DC (r 0.75) and lowers the B/T ratio (r -0.74), whereas blue irradiance raises bottom hardness and B/T (r 0.66 to 0.69). The pooled negative DC-and-hardness correlations are driven by the resin composite difference and are meaningful only within resin composite.

6. Discussion

Why the two transmission methods disagree, and why both are right. Section 1A coupled the exit face of each material to a small cosine corrector; Section 1B suspended the material in air in front of a target larger than the beam. Coupling removes one reflecting interface, worth 3 to 4%, and the small sensor loses light that spreads past it at distance. For Filtek Matrix these two effects explain the 7 to 14 point offset on the single-peak units; on the multi-peak units the violet loss seen only by the spectroradiometer cancels the offset. For EXACLEAR a 5 to 10 mm silicone block acts as a light guide: rays that would spread past the small sensor in air are held in by internal reflection at the block walls, so the coupled sensor reads a gain while the profiler, which captures the whole footprint in both cases, reads the net optical loss. The clinical geometry is closer to 1A on the exit side, where the matrix sits against resin composite, and to 1B on the entry side. Neither number alone describes the mouth; together they bracket it. A third measurement, total radiant power in an integrating sphere at contact, agrees with the profiler within 1% for three units and anchors the profiler calibration (Table M1).

Why the violet loss does not reach the resin composite. Filtek Matrix removes a quarter to two thirds of the violet output of multi-peak units and essentially none of the blue. Camphorquinone, the primary initiator in both resin composites, absorbs blue; the alternative initiator in Vit-l-escence absorbs violet. At 20 seconds, every condition delivered 13 to 23 J/cm², above the response plateau of both resin composites, so the blue that remained was sufficient and the violet that was lost was not needed. Section 2C confirms this directly: delivered irradiance predicts nothing across the 20 groups. Within Vit-l-escence the violet that does arrive raises surface conversion and lowers bottom hardness, because violet is absorbed within the first millimeter and does not reach depth; that is a property of the unit and the resin composite, not of the matrix, and it is present in the air control.

Why the beam is unchanged. Loss through Filtek Matrix in an air path decomposes into about 6% at the two surfaces plus 2 to 4% per mm in the bulk (Tables 4 and 10). The interface term is Fresnel reflection and is independent of position; the bulk term is small. The difference maps of Figure 2 show that what is lost is lost uniformly across the plateau and a fraction reappears just outside the edge, which is the signature of a weak diffuser. Over a 10 mm path, a block of refractive index 1.4 to 1.5 also reduces divergence, which is why D4σ is up to 4% smaller through EXACLEAR than in air at 10 mm. Through Filtek Matrix, D4σ is up to 2% larger because the small amount of light scattered just outside the beam edge counts heavily in this measure, even though the plateau itself does not grow (Table 8, Figure 2). The one unit whose beam changes markedly with distance, PowerCure, does so in air and the matrix neither improves nor worsens it.

What governs the outcome. In every Part 2 analysis resin composite was the dominant term (partial η^2 0.59 to 0.83), the LPU $\times$ resin composite interaction was second, and light path was last. The practical reading is that the choice of resin composite and its match to the emission spectrum of the unit decide conversion and depth of polymerization, and tip distance decides how much light arrives; the matrix material sits below all three.

Single-peak compared with multi-peak. Blue transmission is the same for both types. Violet transmission is not, so total irradiance transmission appears unit dependent (83 to 94% multi-peak, 97 to 104% single-peak in 1A) even though the component that drives camphorquinone is delivered equally. Figure 3 makes the reason visible: the two multi-peak units carry 17 and 25% of their output below 425 nm, the two single-peak units carry 1% or less.

Resin composite comparison. The resin composite with the alternative photoinitiator responds to violet at the surface and pays for it at depth; the camphorquinone resin composite responds to total blue and is indifferent to the spectrum of the unit. Thin-film DC cannot serve as a proxy for depth of polymerization with violet-responsive resin composites, which is why the two outcomes were measured separately here.

7. Claims supported by the data

Candidate statement	Supported	Evidence	Qualifier
Filtek Matrix at up to 2 mm transmits blue light (425 to 550 nm) at 96 to 104% of the value at the same distance in air, with single-peak and multi-peak LPUs	Yes	Table 2	Coupled geometry; four LPUs, one unit each
Filtek Matrix does not alter the beam diameter, shape, or hot-spot distribution of an LPU	Yes	Tables 8, 9; Figures 1, 2	One frame per condition; four LPUs
No increase in exposure time is required when polymerizing through Filtek Matrix at up to 2 mm with Elipar DeepCure-S or VALO X for 20 seconds	Yes	Tables 12 to 15	Two resin composites, 2 mm depth, tip on the matrix; n=5, null finding within the power of the study
Degree of conversion and Knoop hardness of Filtek Supreme Ultra polymerized through Filtek	Yes	Tables 12 to 15	No difference detected; detectable differences 1.5 to 2.1 DC points and 8 to 13

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Matrix do not differ from direct contact			KHN (Section 11); not a formal equivalence test
Filtek Matrix attenuates violet light from multi-peak LPUs	Yes, and should be stated	Table 2; Figure 3	31 to 77% transmitted at 1 to 2 mm; no measurable effect on conversion or hardness at 2 mm and 20 seconds
Filtek Matrix improves or homogenizes the beam of an LPU	Not supported	Table 9; Figure 2	Softening of edges is visible but uniformity metrics change by less than 0.02
Filtek Matrix compensates for tip distance	Not supported	Tables 1, 10; Figure 1	Air distance losses are unchanged by the matrix; PowerCure beam collapse with distance is unaffected
Results apply to the shaped clinical matrix	Not directly	Section 11	Flat 20 × 20 mm plates only

8. Clinical recommendations

- Elipar DeepCure-S: polymerize through Filtek Matrix at 1 to 2 mm with the standard 20-second exposure and the tip on the matrix. No compensation is needed. Blue delivery through the matrix exceeds 1600 mW/cm².
- SmartLite Pro (standard tip): same, inferred from blue transmission (Section 1A); polymerization was not tested with this unit.
- VALO X: same for camphorquinone resin composites. Blue delivery is unchanged (about 900 to 980 mW/cm² through the matrix); violet is reduced to one third to one half. For resin composites that rely on violet-sensitive initiators, a small reduction in surface response is expected but no change in depth of polymerization.
- Bluephase PowerCure: same as VALO X, with a smaller violet reduction, inferred from transmission data; polymerization was not tested with this unit. Because the beam of this unit degrades markedly with distance in air (Figure 1), keeping the tip in contact with the matrix matters more for this unit than for the others.
- Across all LPUs, tip distance is the dominant variable. One centimeter of air costs more irradiance than 2 mm of Filtek Matrix or 10 mm of EXACLEAR.
- EXACLEAR at 5 to 10 mm in contact with the target delivers as much or more irradiance than the equivalent air gap and does not require compensation.

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9. Conclusions

1. Filtek Matrix transmits blue light nearly fully and attenuates violet; only multi-peak units are affected, and only in the violet band.
2. Neither Filtek Matrix nor EXACLEAR alters beam size, shape, uniformity, or hot-spot distribution; both act as weak diffusers.
3. Polymerizing through either material for 20 seconds produced no detectable change in conversion or hardness compared with direct contact for both resin composites and both LPUs; the study was sensitive to a 1 to 2 point loss in conversion.
4. Resin composite chemistry, its match to the emission spectrum of the LPU, and tip distance determine depth of polymerization; the interposed material does not.

10. Deviations from the proposal

Proposed	Executed	Consequence
Integrating sphere, 380 to 520 nm at 1 nm, for transmission through the materials	Integrating sphere used for radiant power and peak wavelength of each LPU at contact (Table M1); transmission through the materials measured with a cosine-corrector spectroradiometer in three bands 350 to 550 nm	Cosine-corrector sensor smaller than beam; values at distance include overfill; exit face coupled. Wavelength-resolved transmission is available as the ratio of through-material to matched-air spectra for each LPU.
Phase 1: 80 measurements	180 measurements (air series at five distances added)	Matched-distance baseline for every material condition.
Beam profile at 8 mm, 5 replicates	0 to 10 mm series, 1 frame each	More conditions, no inferential statistics.
Four LPUs in Phase 3	Two LPUs (DeepCure-S, VALO X)	100 rather than 160 specimens per outcome; the two bracket single-peak and multi-peak and include the unit with the largest violet loss.
DC at the bottom of 2 mm resin composite, three readings averaged	Thin-film DC from a 6-minute kinetic trace, single value at 300 seconds after light initiation	Depth of polymerization carried by hardness only; kinetic traces available for rate analysis.
10 mm diameter molds	5 mm	Disks sit within the beam plateau; no spatial mapping.

Vickers, 50 g, 10 seconds, bottom only, 15 indents, spatial maps	Knoop, 50 gf, 15 seconds, top and bottom, 5 indents, no spatial map	Knoop appropriate for polymer surfaces; bottom-to-top ratio added; hardness maps not produced.
Correlation with beam profile	Correlation with delivered irradiance (Section 2C)	Delivered.
Statistical software: Prism 10	Prism 11.0.0, independently reproduced in Python	No effect.

11. Limitations

One unit of each LPU was tested. Section 1A replicates are repeat readings of the same unit, so statistical significance in Tables 1 to 3 reflects instrument precision rather than between-unit variation; inference is limited to the units tested. Section 1B has one frame per condition. Part 2 used two of four LPUs and one exposure time. Flat printed plates represent a best case for transmission relative to the shaped clinical matrix. Null results have finite power. With $n=5$ and Dunnett comparisons against air ($\alpha=.05$, 80% power), the minimal detectable difference was 1.2 to 2.1 points for DC, 8 to 13 KHN for bottom hardness of Filtek Supreme Ultra, and 18 to 21 KHN for bottom hardness of Vit-l-escence, whose indentation variability was high. The DC null result therefore excludes any meaningful loss of conversion, whereas the hardness null result excludes only large losses, particularly for Vit-l-escence. The spectral response of the profiler is not flat, which biases through-matrix power ratios for multi-peak units. The photoinitiator composition of Vit-l-escence is inferred from its violet response and manufacturer information.

12. Recommended next steps

1. Spectrophotometric transmission of Filtek Matrix and EXACLEAR, 380 to 520 nm, at each thickness, to obtain the attenuation coefficient as a function of wavelength and the violet cutoff.
2. Beam profiling with 405, 420, 450, and 470 nm bandpass filters, five frames per condition, to resolve per-band transmission and the spatial distribution of the violet component through the matrix, with a specimen rotation check to confirm the origin of the striations seen in Figure 1.
3. Optional: hardness with the shaped clinical matrix rather than flat plates, and with 10 mm disks and 15 indents to produce the spatial maps originally proposed.

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4. Extract polymerization rate and time to plateau from the 100 kinetic DC traces already acquired, which delivers the kinetic component of the proposal without new measurements.

13. Data integrity and availability

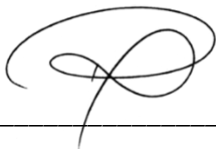
Raw data: TRANSMISSION_STATS.prism;
DC_STATS.prism; MICROHARDNESS_TOP_STATS.prism, MICROHARDNESS_BOT_STATS.prism, MICROHARDNESS_RATIO_STATS.prism (version 2, exclusions applied, with the original values retained in the project archive); 36 BeamGage .bgData frames; 45 integrating sphere exports (nine groups of five readings, March 25 and May 20, 2026, with full 360 to 830 nm spectra); 100 FTIR kinetic acquisitions with exported spectra, ratio traces, and light-activation traces (.DPT), August 10 to 12, 2026; Knoop indentation files. Location: box folder, access controlled by the principal investigator, retained for a minimum of 5 years. The excluded hardness specimens are documented in this report (Appendix B) and are not deleted from the archive. All statistics were computed in Prism 11.0.0 and independently reproduced from the exported data tables in Python 3 (statsmodels, SciPy); the two agree to the reported precision.

14. Conflict of interest

This study was funded by Solventum, which manufactures Filtek Matrix, Filtek Supreme Ultra, and Elipar DeepCure-S. Study design, execution, analysis, and interpretation were performed independently by the investigators.

15. Sign-off

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Signature:  _____ Date: September 12, 2026

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Appendix A. Supplementary statistics

A1. Section 1A, spectral irradiance: two-way ANOVA tables

Table A1. Air distance series, total 350 to 550 nm (LPU $\times$ distance; 4 LPUs $\times$ 5 distances, n=5)

Source	df	F	P	partial η^2
LPU	3	6801.76	<.001	1.00
Distance	4	13244.58	<.001	1.00
LPU $\times$ distance	12	145.77	<.001	0.96
Residual	80			

Table A2. Air distance series, blue 425 to 550 nm

Source	df	F	P	partial η^2
LPU	3	12344.83	<.001	1.00
Distance	4	10898.06	<.001	1.00
LPU $\times$ distance	12	186.90	<.001	0.97
Residual	80			

Table A3. Air distance series, violet 350 to 425 nm

Source	df	F	P	partial η^2
LPU	3	110189.80	<.001	1.00
Distance	4	12364.95	<.001	1.00
LPU $\times$ distance	12	4081.03	<.001	1.00
Residual	80			

Table A4. Filtek Matrix 1 mm and matched air, total 350 to 550 nm (LPU $\times$ material, n=5)

Source	df	F	P	partial η^2
LPU	3	3345.15	<.001	1.00
Material (air or Filtek Matrix)	1	204.29	<.001	0.86
LPU $\times$ material	3	31.91	<.001	0.75
Residual	32			

Table A5. Filtek Matrix 2 mm and matched air, total 350 to 550 nm

Source	df	F	P	partial η^2
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LPU	3	5039.78	<.001	1.00
Material (air or Filtek Matrix)	1	406.85	<.001	0.93
LPU × material	3	165.24	<.001	0.94
Residual	32			

Table A6. EXACLEAR 5 mm and matched air, total 350 to 550 nm

Source	df	F	P	partial η^2
LPU	3	1561.30	<.001	0.99
Material (air or EXACLEAR)	1	64.50	<.001	0.67
LPU × material	3	6.27	.002	0.37
Residual	32			

Table A7. EXACLEAR 10 mm and matched air, total 350 to 550 nm

Source	df	F	P	partial η^2
LPU	3	3291.29	<.001	1.00
Material (air or EXACLEAR)	1	431.09	<.001	0.93
LPU × material	3	90.53	<.001	0.89
Residual	32			

Table A8. Filtek Matrix 1 mm and matched air, blue 425 to 550 nm

Source	df	F	P	partial η^2
LPU	3	5702.79	<.001	1.00
Material (air or Filtek Matrix)	1	16.38	<.001	0.34
LPU × material	3	9.90	<.001	0.48
Residual	32			

Table A9. Filtek Matrix 2 mm and matched air, blue 425 to 550 nm

Source	df	F	P	partial η^2
LPU	3	8671.45	<.001	1.00
Material (air or Filtek Matrix)	1	4.00	.054	0.11
LPU × material	3	26.93	<.001	0.72
Residual	32			

Table A10. EXACLEAR 5 mm and matched air, blue 425 to 550 nm

Source	df	F	P	partial η^2
LPU	3	3161.18	<.001	1.00
Material (air or EXACLEAR)	1	91.37	<.001	0.74
LPU $\times$ material	3	8.53	<.001	0.44
Residual	32			

Table A11. EXACLEAR 10 mm and matched air, blue 425 to 550 nm

Source	df	F	P	partial η^2
LPU	3	6216.89	<.001	1.00
Material (air or EXACLEAR)	1	585.37	<.001	0.95
LPU $\times$ material	3	91.05	<.001	0.90
Residual	32			

Table A12. Filtek Matrix 1 mm and matched air, violet 350 to 425 nm

Source	df	F	P	partial η^2
LPU	3	26658.62	<.001	1.00
Material (air or Filtek Matrix)	1	4273.84	<.001	0.99
LPU $\times$ material	3	2076.89	<.001	0.99
Residual	32			

Table A13. Filtek Matrix 2 mm and matched air, violet 350 to 425 nm

Source	df	F	P	partial η^2
LPU	3	2587.29	<.001	1.00
Material (air or Filtek Matrix)	1	1508.94	<.001	0.98
LPU $\times$ material	3	652.56	<.001	0.98
Residual	32			

Table A14. EXACLEAR 5 mm and matched air, violet 350 to 425 nm

Source	df	F	P	partial η^2
LPU	3	12291.38	<.001	1.00
Material (air or EXACLEAR)	1	3.71	.063	0.10
LPU $\times$ material	3	1.97	.138	0.16
Residual	32			

Table A15. EXACLEAR 10 mm and matched air, violet 350 to 425 nm

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Source	df	F	P	partial η^2
LPU	3	4147.24	<.001	1.00
Material (air or EXACLEAR)	1	0.75	.392	0.02
LPU $\times$ material	3	37.32	<.001	0.78
Residual	32			

Legend, A1 to A15. Each table is a separate two-way ANOVA on absolute irradiance (mW/cm^2 , $n=5$ per cell). Tukey pairwise results for these models are the letters and asterisks in Tables 1 to 3. Partial $\eta^2 = \text{SS}_{\text{effect}} / (\text{SS}_{\text{effect}} + \text{SS}_{\text{residual}})$. Residuals fail normality on the raw scale because irradiance differs 100-fold across units and bands; the same models on $\log_{10}(\text{irradiance})$ give identical conclusions and satisfy the Levene test in all tables but one. The $n=5$ are repeat readings of a single unit of each LPU; significance reflects instrument precision, not between-unit variation.

A2. Section 2A, degree of conversion

Table A16. Three-way ANOVA, DC (LPU $\times$ resin composite $\times$ condition; $N=100$)

Source	df	F	P	partial η^2
Condition	4	2.58	.043	0.11
LPU	1	11.51	.001	0.13
Composite	1	113.17	<.001	0.59
Condition $\times$ LPU	4	1.10	.361	0.05
Condition $\times$ resin composite	4	3.15	.019	0.14
LPU $\times$ resin composite	1	42.63	<.001	0.35
Condition $\times$ LPU $\times$ resin composite	4	0.48	.753	0.02
Residual	80			

Table A17. Two-way ANOVA within Filtek Supreme Ultra, DC (condition $\times$ LPU; $N=50$)

Source	df	F	P	partial η^2
Condition	4	3.44	.017	0.26
LPU	1	4.02	.052	0.09
Condition $\times$ LPU	4	0.37	.827	0.04
Residual	40			

Table A18. Two-way ANOVA within Vit-l-escence, DC (condition $\times$ LPU; $N=50$)

Source	df	F	P	partial η^2
Condition	4	1.96	.119	0.16

LPU	1	63.29	<.001	0.61
Condition × LPU	4	1.44	.237	0.13
Residual	40			

Table A19. Two-way ANOVA within DeepCure-S, DC (condition × resin composite; N=50)

Source	df	F	P	partial η^2
Condition	4	0.85	.500	0.08
Composite	1	9.19	.004	0.19
Condition × resin composite	4	3.04	.028	0.23
Residual	40			

Table A20. Two-way ANOVA within VALO X, DC (condition × resin composite; N=50)

Source	df	F	P	partial η^2
Condition	4	2.68	.045	0.21
Composite	1	136.25	<.001	0.77
Condition × resin composite	4	0.77	.553	0.07
Residual	40			

Table A21. One-way ANOVA (condition) within each LPU and resin composite pairing, with Dunnett comparisons against air, DC

LPU / resin composite	One-way F	One-way P	Dunnett P: FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm
DeepCure-S / Filtek Supreme Ultra	1.79	.170	.999	.169	.274	.317
DeepCure-S / Vit-l-escence	2.08	.122	.756	.086	.399	1.000
VALO X / Filtek Supreme Ultra	1.96	.139	.529	.181	.142	.065
VALO X / Vit-l-escence	1.02	.423	.610	.972	.610	.227

A3. Section 2B, top Knoop hardness

Table A22. Three-way ANOVA, top KHN (LPU × resin composite × condition; N=98)

Source	df	F	P	partial η^2
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Condition	4	0.64	.636	0.03
LPU	1	0.86	.356	0.01
Composite	1	187.56	<.001	0.71
Condition × LPU	4	1.08	.374	0.05
Condition × resin composite	4	1.21	.314	0.06
LPU × resin composite	1	9.16	.003	0.11
Condition × LPU × resin composite	4	0.54	.706	0.03
Residual	78			

Table A23. Two-way ANOVA within Filtek Supreme Ultra, top KHN (condition × LPU; N=48)

Source	df	F	P	partial η^2
Condition	4	1.74	.162	0.15
LPU	1	2.92	.095	0.07
Condition × LPU	4	0.90	.475	0.09
Residual	38			

Table A24. Two-way ANOVA within Vit-l-escence, top KHN (condition × LPU; N=50)

Source	df	F	P	partial η^2
Condition	4	0.51	.727	0.05
LPU	1	6.06	.018	0.13
Condition × LPU	4	0.76	.556	0.07
Residual	40			

Table A25. Two-way ANOVA within DeepCure-S, top KHN (condition × resin composite; N=48)

Source	df	F	P	partial η^2
Condition	4	0.31	.868	0.03
Composite	1	146.70	<.001	0.79
Condition × resin composite	4	0.59	.674	0.06
Residual	38			

Table A26. Two-way ANOVA within VALO X, top KHN (condition × resin composite; N=50)

Source	df	F	P	partial η^2
Condition	4	1.44	.240	0.13
Composite	1	55.52	<.001	0.58

Condition × resin composite	4	1.14	.354	0.10
Residual	40			

Table A27. One-way ANOVA within each LPU and resin composite pairing, with Dunnett comparisons against air, top KHN

LPU / resin composite	One-way F	One-way P	Dunnett P: FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm
DeepCure-S / Filtek Supreme Ultra	0.29	.879	.997	.997	1.000	.896
DeepCure-S / Vit-l-escence	0.53	.717	.902	.999	.731	.819
VALO X / Filtek Supreme Ultra	1.93	.144	.234	.276	.993	.949
VALO X / Vit-l-escence	0.76	.562	.976	.999	.699	.907

A4. Section 2B, bottom Knoop hardness

Table A28. Three-way ANOVA, bottom KHN (LPU × resin composite × condition; N=97)

Source	df	F	P	partial η^2
Condition	4	0.28	.887	0.01
LPU	1	14.70	<.001	0.16
Composite	1	369.97	<.001	0.83
Condition × LPU	4	0.57	.684	0.03
Condition × resin composite	4	1.01	.410	0.05
LPU × resin composite	1	0.14	.709	0.00
Condition × LPU × resin composite	4	0.45	.776	0.02
Residual	77			

Table A29. Two-way ANOVA within Filtek Supreme Ultra, bottom KHN (condition × LPU; N=47)

Source	df	F	P	partial η^2
Condition	4	1.97	.119	0.18
LPU	1	11.65	.002	0.24
Condition × LPU	4	1.00	.418	0.10
Residual	37			

Table A30. Two-way ANOVA within Vit-l-escence, bottom KHN (condition × LPU; N=50)

Source	df	F	P	partial η^2
Condition	4	0.22	.928	0.02
LPU	1	6.19	.017	0.13
Condition × LPU	4	0.36	.838	0.03
Residual	40			

Table A31. Two-way ANOVA within DeepCure-S, bottom KHN (condition × resin composite; N=47)

Source	df	F	P	partial η^2
Condition	4	0.34	.849	0.04
Composite	1	215.66	<.001	0.85
Condition × resin composite	4	1.24	.312	0.12
Residual	37			

Table A32. Two-way ANOVA within VALO X, bottom KHN (condition × resin composite; N=50)

Source	df	F	P	partial η^2
Condition	4	0.49	.742	0.05
Composite	1	166.75	<.001	0.81
Condition × resin composite	4	0.39	.815	0.04
Residual	40			

Table A33. One-way ANOVA within each LPU and resin composite pairing, with Dunnett comparisons against air, bottom KHN

LPU / resin composite	One-way F	One-way P	Dunnett P: FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm
DeepCure-S / Filtek Supreme Ultra	3.16	.041	.999	.456	.631	.231
DeepCure-S / Vit-l-escence	0.35	.839	.998	.763	1.000	1.000
VALO X / Filtek Supreme Ultra	0.94	.462	.269	.855	.906	.999
VALO X / Vit-l-escence	0.24	.915	1.000	1.000	.880	1.000

A5. Section 2B, bottom-to-top hardness ratio

Table A34. Three-way ANOVA, B/T ratio (LPU × resin composite × condition; N=97)

Source	df	F	P	partial η^2
Condition	4	0.45	.769	0.02
LPU	1	24.47	<.001	0.24
Composite	1	119.42	<.001	0.61
Condition × LPU	4	0.20	.936	0.01
Condition × resin composite	4	0.48	.748	0.02
LPU × resin composite	1	10.81	.002	0.12
Condition × LPU × resin composite	4	0.57	.683	0.03
Residual	77			

Table A35. Two-way ANOVA within Filtek Supreme Ultra, B/T ratio (condition × LPU; N=47)

Source	df	F	P	partial η^2
Condition	4	1.49	.226	0.14
LPU	1	2.57	.117	0.06
Condition × LPU	4	0.71	.592	0.07
Residual	37			

Table A36. Two-way ANOVA within Vit-l-escence, B/T ratio (condition × LPU; N=50)

Source	df	F	P	partial η^2
Condition	4	0.18	.947	0.02
LPU	1	22.53	<.001	0.36
Condition × LPU	4	0.30	.874	0.03
Residual	40			

Table A37. Two-way ANOVA within DeepCure-S, B/T ratio (condition × resin composite; N=47)

Source	df	F	P	partial η^2
Condition	4	0.42	.795	0.04
Composite	1	31.57	<.001	0.46
Condition × resin composite	4	0.71	.592	0.07
Residual	37			

Table A38. Two-way ANOVA within VALO X, B/T ratio (condition × resin composite; N=50)

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Source	df	F	P	partial η^2
Condition	4	0.26	.900	0.03
Composite	1	91.42	<.001	0.70
Condition $\times$ resin composite	4	0.39	.811	0.04
Residual	40			

Table A39. One-way ANOVA within each LPU and resin composite pairing, with Dunnett comparisons against air, B/T ratio

LPU / resin composite	One-way F	One-way P	Dunnett P: FM 1 mm	FM 2 mm	EXA 5 mm	EXA 10 mm
DeepCure-S / Filtek Supreme Ultra	2.26	.106	1.000	.597	.379	.623
DeepCure-S / Vit-l-escence	0.37	.827	.664	.708	.905	.945
VALO X / Filtek Supreme Ultra	0.82	.529	1.000	.821	.760	.995
VALO X / Vit-l-escence	0.13	.972	.999	1.000	.996	.977

Legend, A16 to A39. Condition = light path (air contact; Filtek Matrix 1 and 2 mm; EXACLEAR 5 and 10 mm). DC was measured on thin films by FTIR-ATR at 300 seconds after light initiation; hardness on 5 $\times$ 2 mm disks at 24 hours, five Knoop indents per surface at 50 gf with a 15-second dwell. Partial η^2 = SS_effect / (SS_effect + SS_residual). Dunnett P values are two sided, adjusted for four comparisons against the air control within each LPU and resin composite pairing; none is below .05 for any outcome. Hardness analyses exclude unmeasurable surfaces on three DeepCure-S / Filtek Supreme Ultra specimens (FM 2 mm, top and bottom of two specimens; EXACLEAR 5 mm, bottom of one specimen). Tukey letters for the LPU $\times$ resin composite interaction are in Tables 12 to 15 of the main text.

Appendix B. Hardness statistics with the excluded specimens retained

With the excluded DeepCure-S / Filtek Supreme Ultra specimens included (N=100 per outcome): condition P =.32 (top KHN), .78 (bottom KHN), .53 (B/T ratio); no Dunnett comparison against air was significant for any pairing; LPU $\times$ resin composite P =.036 (top), .71 (bottom), .001 (B/T). Cell means for DeepCure-S / Filtek Supreme Ultra bottom KHN were 61.6 $\pm$ 15.4 (FM 2 mm) and 63.0 $\pm$ 14.0 (EXACLEAR 5 mm) compared with 68.0 $\pm$ 1.5 and 69.0 $\pm$ 4.2 after exclusion. No conclusion changes.

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Appendix C. Beam profile processing

Frames were extracted from the HDF5 .bgData containers (BG_DATA/1/DATA), converted to irradiance with the stored power-calibration multiplier and the 12.99 μm effective pixel scale, and pedestal corrected from the four frame corners. The SmartLite Pro 0 mm frame carried a uniform 37 mW/cm^2 pedestal from stray light and was corrected accordingly. Because the SmartLite Pro power calibration was set on that 0 mm frame with the pedestal present, the pedestal-corrected SmartLite Pro frames read 884 mW at contact against 983 mW measured in the integrating sphere; SmartLite Pro absolute values in Tables 5 to 7 and in Figure 1 were therefore rescaled by $983/884 = 1.112$. Transmission ratios and beam geometry are unaffected. The other three units agree with the sphere within 1% without correction. All extraction, calibration, metric computation, and figure generation were performed with custom Python code written by the principal investigator for this study, which reads the native .bgData containers directly so that no image export step intervenes between the camera frames and the reported values. Peak irradiance and the beam centroid were determined from a lightly smoothed copy of each frame; all other metrics were computed on the unsmoothed data. Per-frame metrics (pedestal, power in aperture, percent in aperture, peak, $D4\sigma$, mean irradiance within $D4\sigma$, area and mean irradiance above 13.5% of peak, plateau CV) are in beam_metrics_raw.csv, and the per-tile irradiance arrays behind Figure 1 are retained and can be supplied for redrawing the figures. The plateau CV in Table 9 is the coefficient of variation of irradiance over pixels above 50% of peak, computed by the custom code; it is not the BeamGage Plateau Uniformity result (ISO 13694, FWHM of the irradiance histogram divided by peak), which is retained in the .bgData files. Figures 1 to 4 are supplied as Figure_1_beam_panel_common_scale.png, Figure_2_difference_maps.png, Figure_3_emission_spectra.png, and Figure_4_spectral_transmission.png. Figure 1 uses a perceptually ordered rainbow scale so that the full 0 to 1800 mW/cm^2 range is legible in a single panel; Figures 3 and 4 are in the Augusta University palette.

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Appendix D. Abstracts for submission to AADOCR 2027

D1. Spectral Irradiance Transmission of Four Light-Curing Units Through a Clear Polymer Matrix and a Clear Silicone Index

Zoella Luther, Navid Khorashahi, Tyler Diamond, Frederick Rueggeberg, Rafael Rocha Pacheco.

Objectives: The purpose of this study was to quantify the irradiance delivered by four light-curing units (LCUs) through a custom 3D-printed clear polymer matrix and a clear vinyl polysiloxane index, relative to the same distance in air, across the violet (350 to 425 nm) and blue (425 to 550 nm) bands.

Methods: Four LCUs (Elipar DeepCure-S, Solventum; SmartLite Pro, Dentsply Sirona; VALO X, Ultradent; Bluephase PowerCure, Ivoclar) were measured with a spectroradiometer (STS, Ocean Optics) fitted with a cosine corrector (n=5). The tip was placed directly on the cosine corrector at 0 mm; air gaps of 1, 2, 5 and 10 mm were set with spacers on an optical breadboard. Flat 20 x 20 mm specimens of the matrix (Filtek Matrix, Solventum; 1 and 2 mm) and index (EXACLEAR, GC; 5 and 10 mm) were held in direct contact with the cosine corrector. Data were analyzed by two-way ANOVA (LCU x path condition) with Tukey post hoc tests ($\alpha=0.05$) for each band and thickness.

Results: In air, irradiance at 10 mm fell to 30 to 46% of contact for all LCUs. Through the matrix, total (350 to 550 nm) transmission relative to matched air was 89 to 100% at 1 mm and 83 to 104% at 2 mm, with the LCU x condition interaction significant ($p<0.001$). Blue-band transmission was 96 to 104% for all LCUs. Violet-band transmission was 92 to 102% for DeepCure-S and SmartLite Pro, but 56% and 31% for VALO X and 77% and 58% for PowerCure at 1 and 2 mm, respectively ($p<0.001$). Through the index, transmission equaled or exceeded matched air: 101 to 108% at 5 mm and 102 to 135% at 10 mm, significant for 5 of 8 LCU and thickness combinations ($p<0.05$).

Conclusions: In a coupled geometry, the clear polymer matrix transmitted blue light almost fully but attenuated the violet output of multi-peak LCUs substantially, and the silicone index delivered more irradiance than an equivalent air gap. In summary, losses were wavelength- and LCU-dependent, and air distance reduced irradiance far more than either material.

Keywords: light-curing units, irradiance, clear matrix, multi-peak, transmission. Supported by Solventum.

D2. Beam Profile of Four Light-Curing Units Through Light-Transmitting Matrix Materials

Rafael Rocha Pacheco, Tyler Diamond, Zoella Luther, Navid Khorashahi, Frederick Rueggeberg.

Objectives: The purpose of this study was to describe how a custom 3D-printed clear polymer matrix and a clear vinyl polysiloxane index altered the beam profile of four light-curing units (LCUs) compared with the same distance in air.

Methods: Four LCUs (Elipar DeepCure-S, Solventum; SmartLite Pro, Dentsply Sirona; VALO X, Ultradent; Bluephase PowerCure, Ivoclar) were imaged on a laser beam profiler (SP932U, Ophir-Spiricon) at 0, 1, 2, 5 and 10 mm in air, and through flat 20 x 20 mm specimens of the matrix (Filtek Matrix, Solventum; 1 and 2 mm) and index (EXACLEAR, GC; 5 and 10 mm) placed in the air path between tip and target. Calibrated frames were extracted from the native files, pedestal-corrected, displayed on a common irradiance scale, and centroid-registered. Material-minus-air difference maps and orthogonal profiles were generated at matched distances; $D4\sigma$ diameter and plateau coefficient of variation (CV) were computed as descriptors.

Results: At contact each LCU showed a distinct signature: a fiber-rod mosaic in a round outline (DeepCure-S), a scalloped lens outline with a central peak (PowerCure), a mottled disc with discrete hot regions (SmartLite Pro), and a large flat disc with a sharp rim (VALO X). Through the matrix each beam appeared as a mildly softened version of its air counterpart: edges blurred, the DeepCure-S mosaic washed out, hot-spot positions were unchanged, and a fine periodic striation consistent with the printed surface of the matrix was superimposed on SmartLite Pro and VALO X. Difference maps showed uniform loss across the plateau and a narrow gain just outside the beam edge on all LCUs, indicating scatter-driven redistribution rather than selective absorption. $D4\sigma$ changed by less than 5% through either material, up to 2% wider through the matrix and up to 4% narrower through the index at 10 mm. PowerCure alone transitioned from a top-hat to a near-Gaussian profile by 5 mm in air (CV 0.147 to 0.185), and neither material altered that transition.

Conclusions: Both materials acted as weak diffusers, preserving each LCU's beam geometry and hot-spot map while softening edges. In summary, the materials did not change where light was delivered, and they did not correct LCUs whose beams degraded with distance.

Keywords: light-curing units, beam profile, clear matrix, beam homogeneity, irradiance distribution. Supported by Solventum.

D3. Degree of Conversion of Resin Composites Cured Through a Clear Polymer Matrix and a Clear Silicone Index

Navid Khorashahi, Zoella Luther, Tyler Diamond, Frederick Rueggeberg, Rafael Rocha Pacheco.

Objectives: The purpose of this study was to determine whether curing through a custom 3D-printed clear polymer matrix or a clear vinyl polysiloxane index changed the degree of conversion (DC) of two resin resin composites polymerized with a single-peak or a multi-peak light-curing unit (LCU).

Methods: Thin films of Filtek Supreme Ultra (A2 Body, Solventum) and Vit-l-escence (Universal A2, Ultradent) were light-cured for 20 s with Elipar DeepCure-S (Solventum; single-peak) or VALO X (Ultradent; multi-peak) under five conditions: tip in contact (air), through flat 20 x 20 mm specimens of the matrix (Filtek Matrix, Solventum) at 1 and 2 mm, or through the index (EXACLEAR, GC) at 5 and 10 mm, with the LCU tip placed directly on the specimen (n=5). DC was measured by FTIR-ATR (Invenio R, Bruker) 300 s after light initiation from the 1636/1608 cm^{-1} C=C peak ratio, and analyzed by three-way ANOVA (LCU x resin composite x condition) with Tukey post hoc tests and Dunnett comparisons against air ($\alpha=0.05$).

Results: DC ranged from 52.5% to 55.9% across all 20 groups. Path condition had a small main effect ($p=0.043$) that no pairwise comparison isolated and did not interact with LCU ($p=0.36$); no matrix or index condition differed from air for any LCU and resin composite combination (Dunnett, all $p\geq 0.065$). Composite and the LCU x resin composite interaction were significant ($p<0.001$): VALO X produced higher DC of Vit-l-escence than DeepCure-S at every condition (55.3 to 55.9% vs 53.7 to 54.8%), including through 2 mm of matrix, whereas Filtek Supreme Ultra converted equally with either LCU (54.0% in air).

Conclusions: Curing through the clear polymer matrix at clinically relevant thickness, or through the silicone index up to 10 mm, did not change DC compared with direct contact curing for either resin composite or LCU. In summary, conversion was governed by resin composite chemistry and its match to the LCU emission spectrum, not by the interposed material.

Keywords: degree of conversion, clear matrix, light-curing units, resin resin composite, FTIR. Supported by Solventum.

D4. Knoop Hardness and Bottom-to-Top Ratio of Resin Composites Cured Through a Clear Polymer Matrix and a Clear Silicone Index

Tyler Diamond, Zoella Luther, Mary Hood, Navid Khorashahi, Frederick Rueggeberg, Rafael Rocha Pacheco.

Objectives: The purpose of this study was to determine whether curing through a custom 3D-printed clear polymer matrix or a clear vinyl polysiloxane index changed the top and bottom Knoop hardness (KHN) and bottom-to-top (B/T) hardness ratio of two resin resin composites polymerized with a single-peak or a multi-peak light-curing unit (LCU).

Methods: Discs (5 mm x 2.0 mm) of Filtek Supreme Ultra (A2 Body, Solventum) and Vit-l-escence (Universal A2, Ultradent) were packed into 3D-printed tooth-shaded resin molds between Mylar strips and light-cured for 20 s with Elipar DeepCure-S (Solventum; single-peak) or VALO X (Ultradent; multi-peak) under five conditions: tip in contact (air), through flat 20 x 20 mm specimens of the matrix (Filtek Matrix, Solventum) at 1 and 2 mm, or through the index (EXACLEAR, GC) at 5 and 10 mm, with the LCU tip placed directly on the specimen (n=5; unmeasurable indentations on three specimens were excluded, leaving n=3 to 5 per surface). After 24 h dry, dark storage at 37 °C, five Knoop indentations per surface were made on top and bottom (HM-220, Mitutoyo; 50 gf, 15 s dwell). Top KHN, bottom KHN and B/T ratio were analyzed by three-way ANOVA (LCU x resin composite x condition) with Tukey post hoc tests and Dunnett comparisons against air ($\alpha=0.05$).

Results: Path condition was not significant for top KHN ($p=0.64$), bottom KHN ($p=0.89$) or B/T ratio ($p=0.77$), did not interact with LCU or resin composite, and no matrix or index condition differed from air for any LCU and resin composite combination. Composite dominated every outcome ($p<0.001$): Filtek Supreme Ultra averaged 72 (top) and 69 (bottom) KHN with B/T 0.89 to 1.01, versus 56 and 40 KHN with B/T 0.61 to 0.84 for Vit-l-escence. The LCU x resin composite interaction was significant for top KHN ($p=0.003$) and B/T ratio ($p=0.002$): with Vit-l-escence, VALO X produced harder top surfaces (58 vs 54 KHN) but softer bottoms (37 vs 44 KHN) than DeepCure-S, giving B/T ratios of 0.61 to 0.67 versus 0.76 to 0.84.

Conclusions: Curing through the clear polymer matrix or the silicone index up to 10 mm did not reduce surface or bottom hardness or the B/T ratio of either resin composite. In summary, hardness at 2 mm was governed by the resin composite and by the LCU's blue irradiance rather than by the interposed material.

Keywords: Knoop hardness, clear matrix, light-curing units, resin resin composite, depth of cure. Supported by Solventum.